

Feature

Ultra-low power start-up

- Cold start from 275 mV input voltage and 3 μ W input power (typical).

Very efficient energy extraction

- Open-circuit voltage sensing for Maximum Power Point Tracking (MPPT).
- Selectable open-circuit voltage ratios from 60% to 90% or fixed impedance.
- Programmable MPPT sensing period.
- MPPT voltage operation range from 100 mV to 4.5 V.

Adaptive and smart energy management

- DCDC switches automatically between boost, buck-boost and buck operation, according to input and output voltages, to maximize energy transfer.
- DCDC automatically handles multiple inputs and outputs:
 - Keeps the internal supply and the load voltages regulated while storing the excess of energy in the storage element.
 - The storage element can be used as an input to keep the load and internal voltages regulated when no energy is available on the source.

Load supply voltage

- Selectable load voltage from 1.2 V to 3.3 V.
- Current drive capability: 30 mA in low power mode, 60 mA in high power mode.

Battery protection features

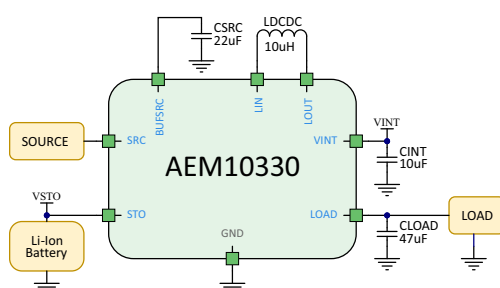
- Selectable over-charge and over-discharge protection for any type of rechargeable battery or (super-)capacitor.
- Fast super-capacitor charging.
- Dual cell super-capacitor balancing circuit.

Smallest footprint, smallest BOM

- Only four external components are required.
 - One 10 μ H inductor.
 - Three capacitors: 10 μ F, 22 μ F and 47 μ F.

Applications

Smart home	Industrial sensor
Smart building	Retail
Edge IoT	PC accessories



Description

The AEM10330 is an integrated energy management circuit that extracts DC power from an ambient energy harvesting source to simultaneously supply an application and store energy in a storage element. The AEM10330 allows to extend battery lifetime and ultimately eliminates the primary energy storage element in a large range of applications.

Thanks to its Maximum Power Point Tracking system, the AEM10330 extracts the maximum energy available from the source. It integrates an ultra-low power DCDC converter which operates with input voltages ranging from 100 mV to 4.5 V.

At start-up, user can choose between charging the storage element first to accumulate energy before the application circuit starts, or charge the load decoupling capacitor first to start the application circuit faster.

With its unique cold start circuit, the AEM10330 can start harvesting with an input voltage as low as 275 mV and from an input power of 3 μ W. The preset protection levels determine the storage element voltages protection thresholds to avoid over-charging and over-discharging the storage element and thus, avoiding damaging it. Those are set through configuration pins. Moreover, custom threshold voltages can be obtained at the expense of a few configuration resistors.

The load voltage can be selected to cover most application needs, with a maximum available load current of 60 mA.

The chip integrates all active elements for powering a typical wireless sensor. Only three capacitors and one inductor are required.

Device Information

Part Number	Package	Body size
10AEM10330J0000	QFN 40-pin	5x5mm

Evaluation Board

Part Number
2AAEM10330J001

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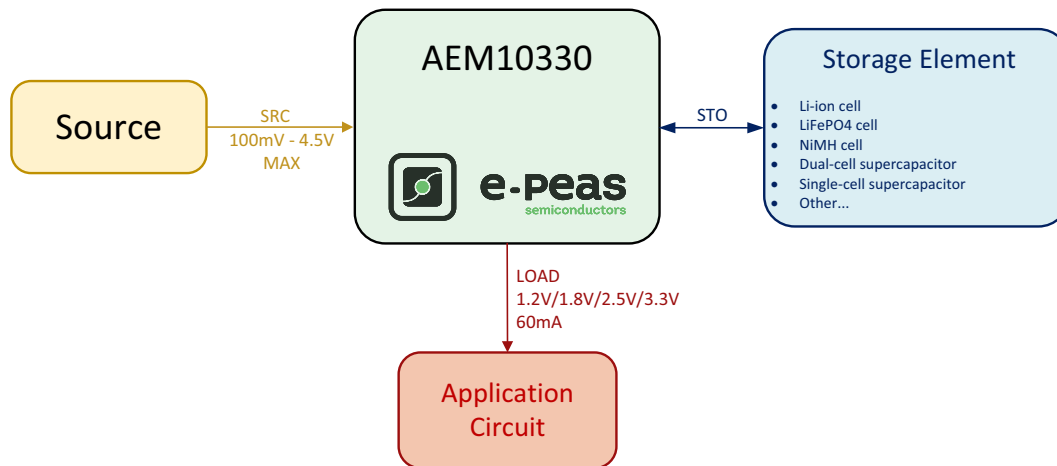


Figure 1: Simplified Schematic View

1. Introduction

The AEM10330 is a full-featured energy efficient power management circuit able to harvest energy from an energy source (connected to **SRC**) to supply an application circuit (connected to **LOAD**) and use any excess of energy to charge a storage element (connected to **STO**). This is done with a minimal bill of material: only capacitors and one inductor are needed for a basic setup.

The heart of the AEM10330 is a regulated switching DCDC converter with high power conversion efficiency.

At first start-up, as soon as a required cold-start voltage of 275 mV and a sparse amount of power of at least 3 μ W is available at the source, the AEM10330 coldstarts. After the cold start, the AEM extracts the power available from the source if the working input voltage is at least 100 mV.

Through four configuration pins (**STO_CFG[3:0]**), the user can select a specific operating mode out of 15 modes that cover most application requirements without any dedicated external component. Those operating modes define the protection levels of the storage element. If none of those 15 modes fit the user's storage element, a custom mode is also available to allow the user to define a mode with custom specifications.

Status pins **ST_STO**, **ST_STO_RDY** and **ST_STO_OVDIS** provide information about the voltage levels of the storage element.

ST_STO is HIGH when the voltage of the storage element V_{STO} is above V_{CHRDY} and is reset when the voltage drops below V_{OVDIS} .

ST_STO_RDY is HIGH when V_{STO} is above V_{CHRDY} , and reset when V_{STO} drops below V_{CHRDY} .

ST_STO_OVDIS is HIGH when V_{STO} drops below V_{OVDIS} and reset when V_{STO} is above V_{OVDIS} .

ST_LOAD is HIGH when the load voltage V_{LOAD} rises above $V_{LOAD,TYP}$, and is reset when V_{LOAD} drops below $V_{LOAD,MIN}$.

The Maximum Power Point (MPP) ratio is configurable thanks to three configuration pins (**R_MPP[2:0]**) and ensures an optimum biasing of the harvester to maximize power extraction. Depending on the harvester, it is possible to adapt the timings of the MPP evaluations with the two configuration pins (**T_MPP[1:0]**) that sets the periodicity and the duration of the MPP evaluation.

Once started, if at any time the load requires more power than can be harvested from the energy source, the AEM10330 automatically uses the storage element to keep the load supplied.

The AEM10330's DCDC converter can work in two modes: **LOW POWER MODE** and **HIGH POWER MODE**, each one of these being optimized for a power range on **SRC** and **LOAD**.

The charging of the storage element can be prevented by pulling **EN_STO_CH** to GND, typically to protect the storage element if the temperature is too low/high to safely charge it.

The AEM10330 also implements a **SLEEP STATE**, which reduces the quiescent current to avoid wasting the energy stored on the storage element when **EN_SLEEP** is HIGH.

At start-up, user can choose to prioritize either starting the application circuit connected on **LOAD**, or charging the storage element connected on **STO**. This is set by the **STO_PRIO** pin.

2. Pin Configuration and Functions

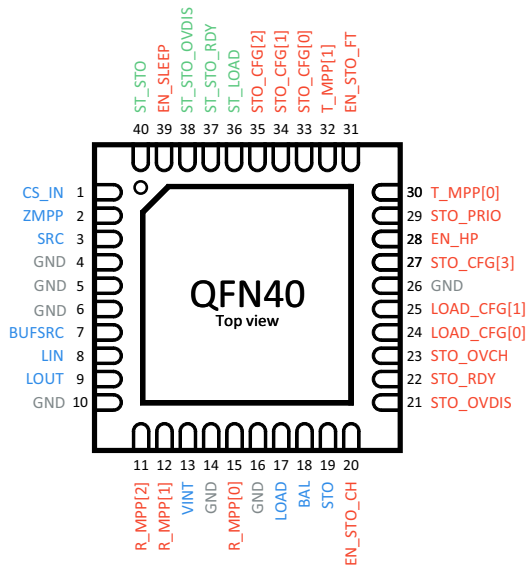


Figure 2: Pinout Diagram QFN 40-pin

NAME	PIN NUMBER	FUNCTION
Power Pins		
CS_IN	1	Input to connect an external cold start circuit.
ZMPP	2	Used for the configuration of the ZMPP (optional). Must be left floating if not used.
SRC	3	Connection to the harvested energy source.
BUFSRC	7	Connection to an external capacitor buffering the DCDC converter input.
LIN	8	DCDC inductance connection.
LOUT	9	DCDC inductance connection.
VINT	13	Internal voltage supply.
LOAD	17	Output voltage to supply an application load.
BAL	18	Connection to mid-point of a dual-cell supercapacitor (optional). Must be connected to GND if not used.
STO	19	Connection to the energy storage element - battery or (super-)capacitor. Cannot be left floating. Must be connected to a minimum capacitance of 100 μ F or to a rechargeable battery.

Table 1: Power and Status Pins



NAME	PIN NUMBER	LOGIC LEVEL		FUNCTION
		LOW	HIGH	
Configuration Pins				
R_MPP[0]	15	GND	VINT	Used to configure the MPPT ratio. Read as HIGH if left floating.
R_MPP[1]	12			
R_MPP[2]	11			
T_MPP[0]	30	GND	VINT	Used to configure the MPPT timings. Read as HIGH if left floating.
T_MPP[1]	32			
LOAD_CFG[0]	24	GND	VINT	Used to configure the LOAD output voltage V_{LOAD} . Read as HIGH if left floating.
LOAD_CFG[1]	25			
STO_CFG[0]	33	GND	VINT	Used to configure the storage element protection thresholds V_{OVDIS} , V_{CHRDY} and V_{OVCH} . Read as HIGH if left floating.
STO_CFG[1]	34			
STO_CFG[2]	35			
STO_CFG[3]	27			
STO_PRIO	29	GND	VINT	Used to configure the charge priority at start-up. - HIGH: storage element (STO) has highest priority at start-up. - LOW: load (LOAD) has highest priority at start-up. Read as HIGH if left floating.
STO_OVCH	23	Analog pins		Used to configure the storage element protection thresholds (V_{OVDIS} , V_{CHRDY} and V_{OVCH}) when STO_CFG[3:0] are set to custom mode (optional). Must be left floating if not used.
STO_RDY	22			
STO_OVDIS	21			
EN_SLEEP	39	GND	LOAD	Used to configure the sleep mode. - HIGH: SLEEP STATE enabled. - LOW: SLEEP STATE disabled. Cannot be left floating.
EN_STO_FT	31	GND	VINT	Used to configure the storage element feed-through feature. - HIGH: allows charges to flow directly from SRC to STO when SRC is above 5V. - LOW: normal operation. Read as HIGH if left floating.
EN_STO_CH	20	GND	LOAD	Used to configure the charge of the storage element. - HIGH: storage element charge enabled. - LOW: storage element charge disabled. Read as HIGH if left floating.
EN_HP	28	GND	VINT	Used to configure the high power mode. - HIGH: HIGH POWER MODE enabled. - LOW: HIGH POWER MODE disabled. Read as HIGH if left floating.

Table 2: Configuration and Ground Pins

Status Pins				
ST_LOAD	36	GND	LOAD	Logic output. - HIGH when the LOAD voltage V_{LOAD} rises above the $V_{LOAD,TYP}$ threshold. - LOW when V_{LOAD} drops below $V_{LOAD,MIN}$ threshold.
ST_STO_RDY	37	GND	LOAD	Logic output. - HIGH when V_{STO} is above V_{CHRDY} . - LOW when V_{STO} drops below V_{CHRDY} .
ST_STO_OVDIS	38	GND	LOAD	Logic output. - HIGH when the AEM10330 is in SHUTDOWN STATE . - LOW when in any other state.
ST_STO	40	GND	LOAD	Logic output. - HIGH when V_{STO} rises above V_{CHRDY} . - LOW when V_{STO} drops below V_{OVDIS} .
Other				
GND	4, 5, 6, 10, 14, 16, 26			Ground connection, best possible connection to PCB ground plane.
	Exposed pad			

Table 2: Configuration and Ground Pins

3. Specifications

3.1. Absolute Maximum Ratings

Parameter		Min	Max	Unit
Operating junction temperature T_J		-40	85	°C
Storage temperature T_{stg}		-65	150	°C
Input voltage	LOAD, STO, SRC, BUFSRC, LIN, LOUT, ZMPP, BAL, CS_IN, EN_SLEEP, EN_STO_CH	-0.3	5.5	V
	VINT, T_MPP[1:0], R_MPP[2:0], LOAD_CFG[1:0], STO_CFG[3:0], STO_PRIO, STO_OVCH, STO_OVDIS, STO_RDY, EN_HP	-0.3	2.75	V

Table 3: Absolute Maximum Ratings

3.2. ESD Ratings

Parameter		Value	Unit
Electrostatic discharge V_{ESD}	Human-Body Model (HBM) according to JEDEC JS001	2000	V
	Charged-Device Model (CDM) according to JEDEC JS002	500	V

Table 4: ESD ratings

ESD CAUTION	
	ESD (ELECTROSTATIC DISCHARGE) SENSITIVE DEVICE These devices have limited built-in ESD protection and damage may thus occur on devices subjected to high-energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality

3.3. Thermal Resistance

Package	θ_{JA}	θ_{JC}	Unit
QFN 40-pin 5x5mm	50	5	°C/W

Table 5: Thermal Resistance



3.4. Typical Electrical Characteristics at 25 °C

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Power Conversion						
P _{SRC,CS}	Minimum source power required for cold start	V _{STO} > V _{CHRDY}		3		μW
		V _{STO} < V _{CHRDY}		6		μW
V _{SRC,CS}	Minimum source input voltage required for cold start			275		mV
V _{SRC}	Input voltage of the energy source	During cold start	V _{SRC,CS}		4.5	V
		After cold start	0.1		4.5	V
R _{MPPT}	MPP Tracker module V _{MPP} / V _{OC} ratio.	See Table 11	60, 65, 70, 75, 80, 85 or 90, depending on R _{MPP} [2:0] configuration			%
Timing						
T _{MPPT,WAIT}	Wait time before V _{OC} measurement begins during MPP evaluation (see Section 5.4).		50% of Table 12		200% of Table 12	ms
T _{MPPT,MEASURE}	Duration of V _{OC} measurement during MPP evaluation.			0.82		ms
T _{MPPT,PERIOD}	Time between two MPP evaluations		50% of Table 12		200% of Table 12	s
T _{GPIO,MON}	GPIO monitoring rate (except EN_SLEEP and EN_STO_CH ¹)			1.3		s
T _{STO,MON}	STO monitoring rate	In WAKE-UP STATE, START STATE, SUPPLY STATE, and SHUTDOWN STATE.		167		μs
		In SLEEP STATE.		667		μs
T _{LOAD,MON}	LOAD monitoring rate	In WAKE-UP STATE, START STATE, SUPPLY STATE, and SHUTDOWN STATE.		167		μs
		In SLEEP STATE.		667		μs
T _{EN_SLEEP,MON}	EN_SLEEP pin monitoring rate			167		μs
Storage element						
V _{OVCH}	Maximum voltage accepted on the storage element before disabling its charging	See Table 9	Depends on STO_CFG[3:0] configuration			V
V _{CHRDY}	Minimum voltage required on the storage element before asserting ST_STO					V
V _{OVDIS}	Minimum voltage accepted on the storage element before entering into SHUTDOWN STATE and resetting ST_STO					V
V _{BAL,MIN}	Minimum voltage required on the BAL pin for the balancing circuit to operate			0.9		V

Table 6: Typical Electrical Characteristics



Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Load Output Voltage						
I _{LOAD,MAX}	LOAD current drive capability (EN_HP = H)	V _{LOAD} = 1.8V V _{STO} > 1.6V		60		mA
		V _{LOAD} = 2.5V V _{STO} > 1.6V		60		
		V _{LOAD} = 3.3V V _{STO} > 1.8V		60		
V _{LOAD}	Output voltage	See Table 10	Depends on LOAD_CFG[1:0] configuration			V
Internal supply & Quiescent Current						
V _{VINT}	Internal voltage supply			2.2		V
I _Q	Quiescent current on STO	V _{STO} = 3.7 V V _{LOAD} = 2.5 V EN_SLEEP = L EN_HP = L		875		nA
I _{Q,EN_STO_CH}	Quiescent current on VINT when EN_STO_CH is pulled down to GND			1.9		μA
Symbol	Logic Level		LOW		HIGH	
Logic output pins						
ST_STO	Logic output levels on the status STO pin		GND		V _{STO}	
ST_LOAD	Logic output levels on the status LOAD pin		GND		V _{LOAD}	
ST_STO_RDY	Logic output levels on the status STO_READY pin		GND		V _{LOAD}	
ST_STO_OVDIS	Logic output levels on the status STO_OVDIS pin		GND		V _{LOAD}	

Table 6: Typical Electrical Characteristics

1. EN_STO_CH pin is monitored continuously.

3.5. Recommended Operating Conditions

Symbol	Parameter	Min	Typ	Max	Unit
External Components					
L_{DCDC}	Inductor of the DCDC converter		10		μH
C_{SRC}	Capacitor decoupling the SRC terminal	13 ¹	22		μF
C_{INT}	Capacitor decoupling the VINT terminal	5 ¹	10		μF
C_{LOAD}	Capacitor decoupling the LOAD terminal	13 ¹	47		μF
C_{STO}	Optional - Capacitor on STO if no storage element is connected (see Section 6.9.1)	100 ¹			μF
R_{ZMPP}	Optional - Used for the configuration of the ZMPP tracking function	10	Section 6.7	100K	Ω
STO_OVCH	Configuration of V_{OVCH} in custom mode	1	Section 6.2	100	$M\Omega$
STO_OVDIS	Configuration of V_{OVDIS} in custom mode				
STO_RDY	Configuration of V_{CHRDY} in custom mode				

Table 7: Recommended Operation Conditions

1. Consider all component tolerance and deratings. Typically, DC-bias derating has a major impact on capacitance on ceramic capacitors.

3.6. Typical Characteristics

3.6.1. DCDC Conversion Efficiency From SRC to STO in Low Power Mode

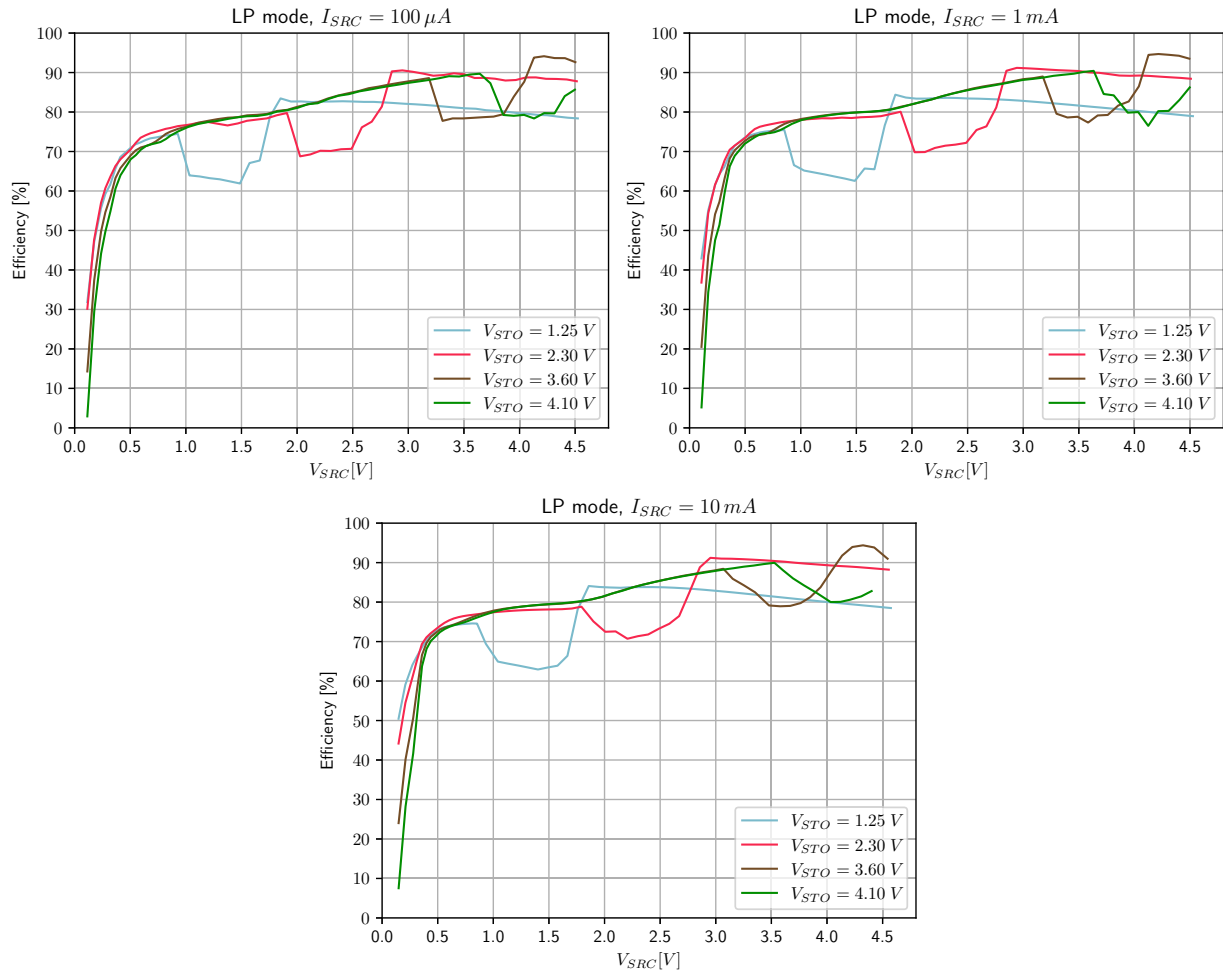


Figure 3: DCDC efficiency from SRC to STO in low power mode



3.6.2. DCDC Conversion Efficiency From SRC to STO in High Power Mode

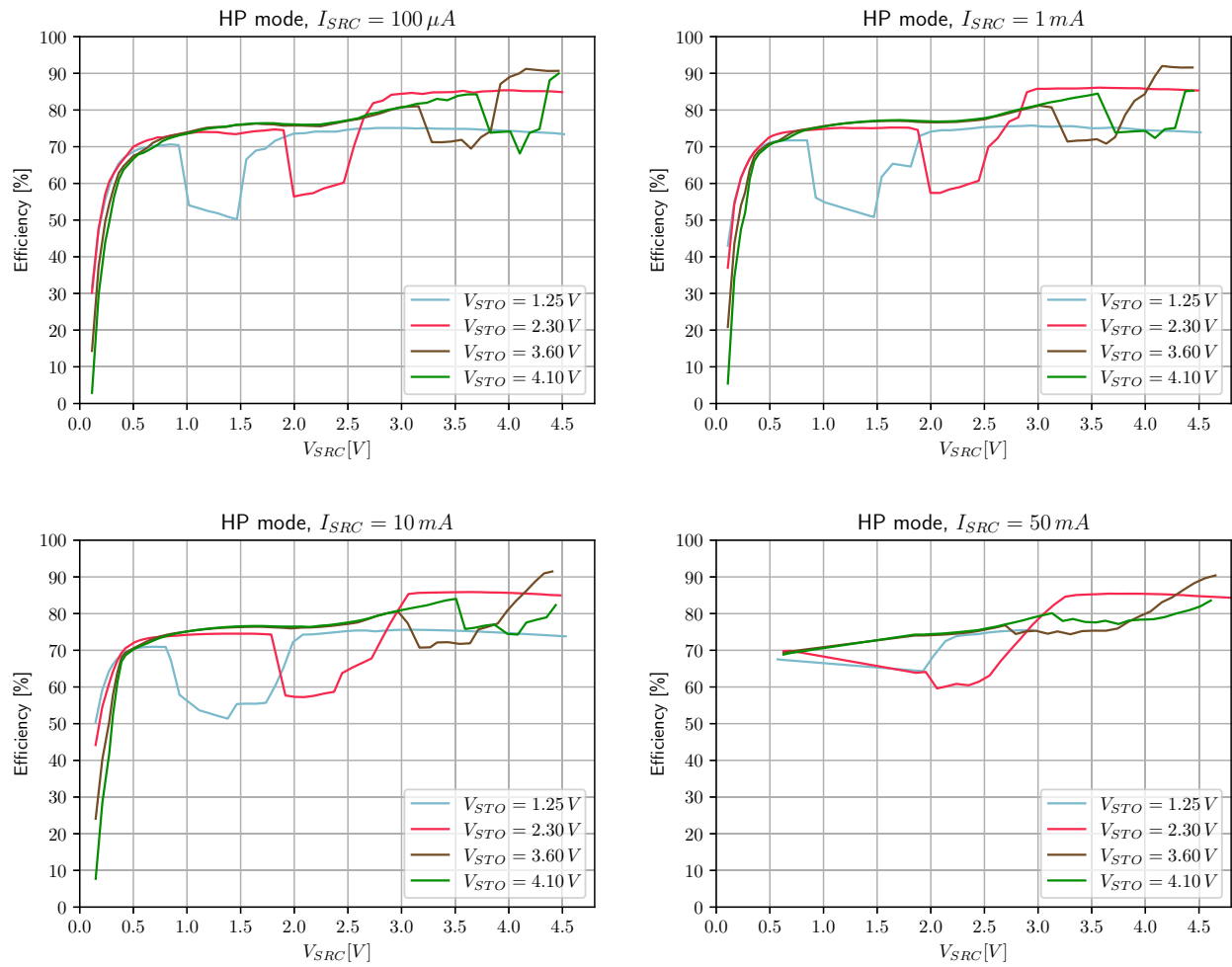


Figure 4: DCDC efficiency from SRC to STO in high power mode

3.6.3. DCDC Conversion Efficiency From STO to LOAD in Low Power Mode

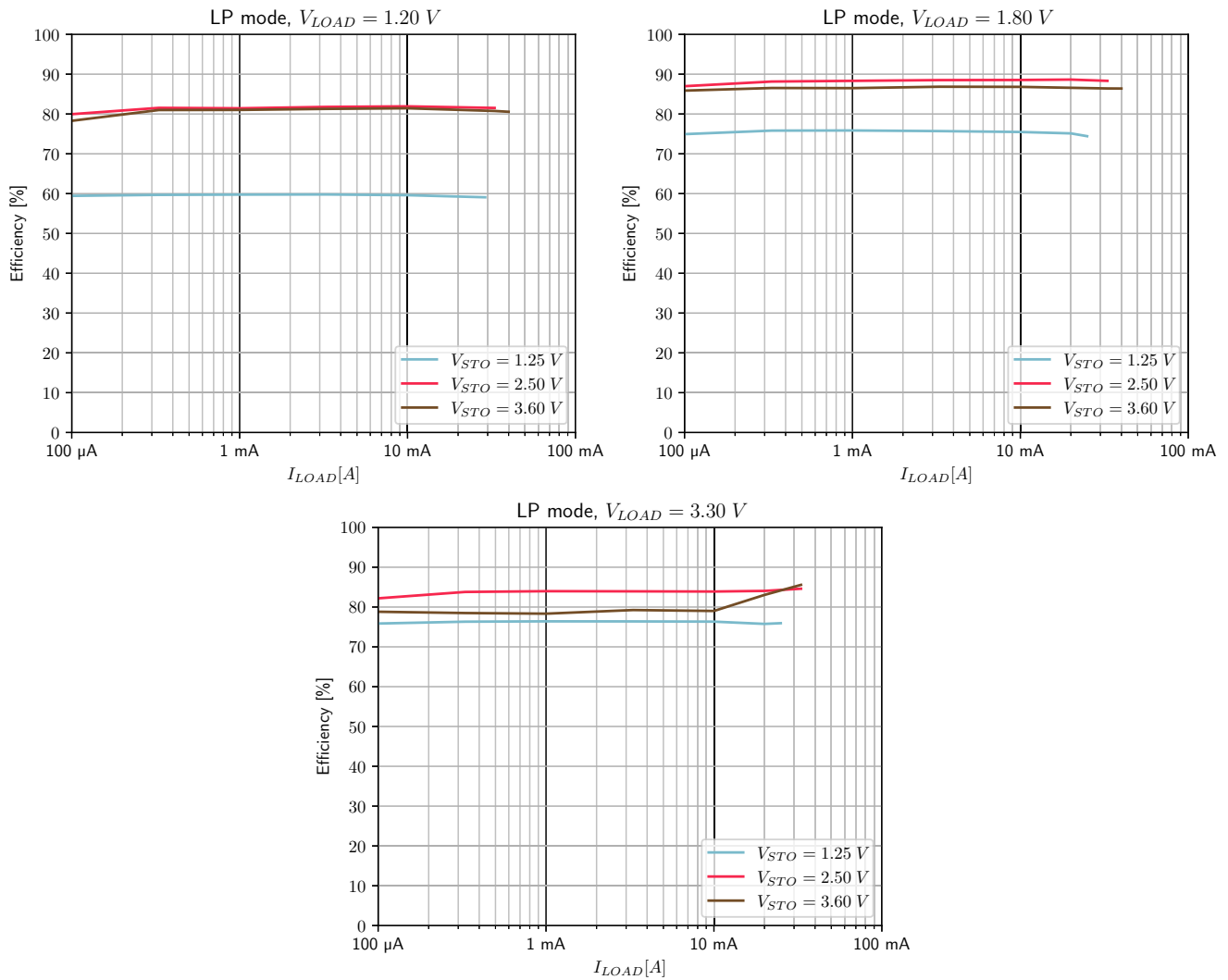


Figure 5: DCDC efficiency from STO to LOAD in low power mode



3.6.4. DCDC Conversion Efficiency From STO to LOAD in High Power Mode

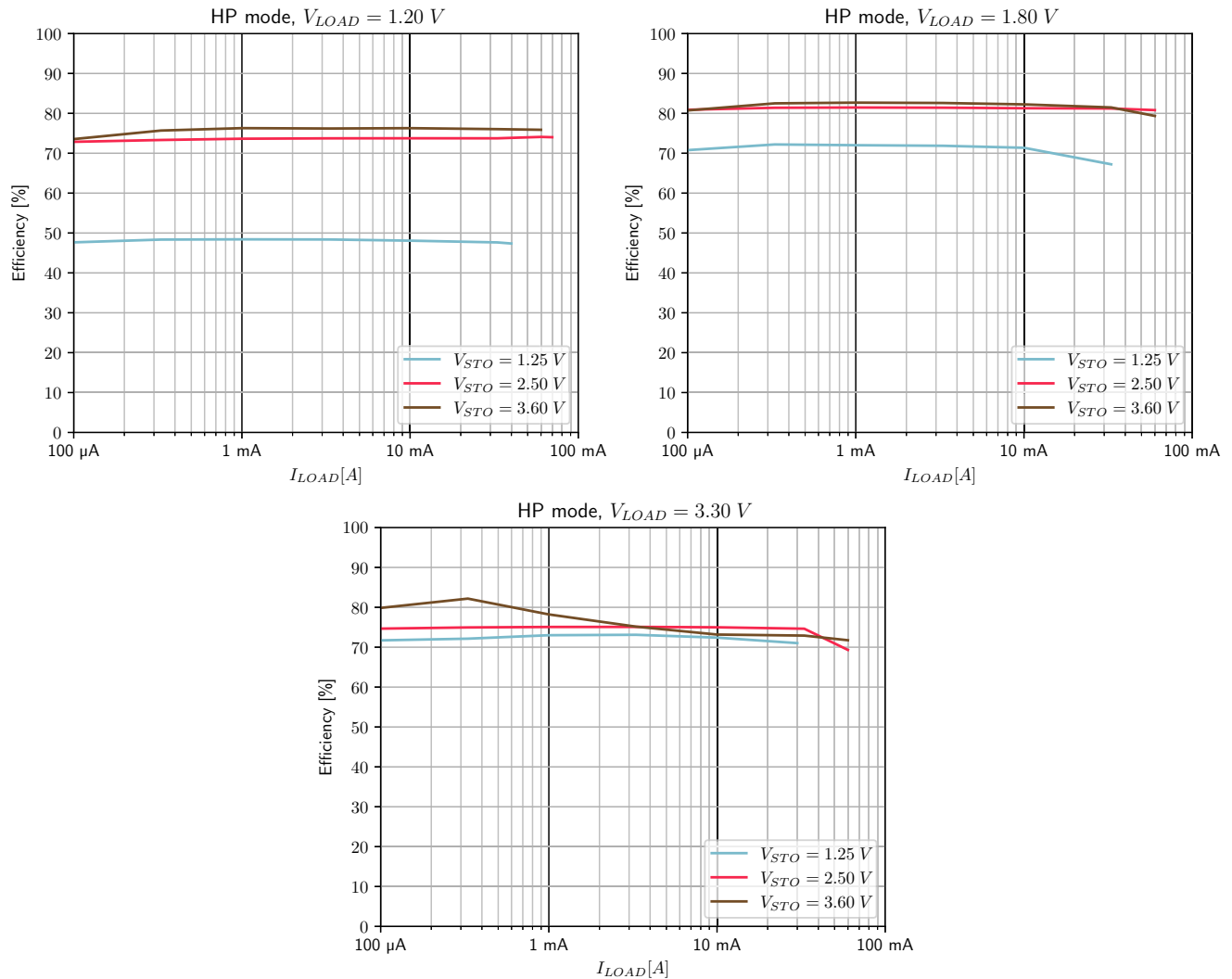


Figure 6: DCDC efficiency from STO to LOAD in high power mode

3.6.5. Maximum LOAD Current

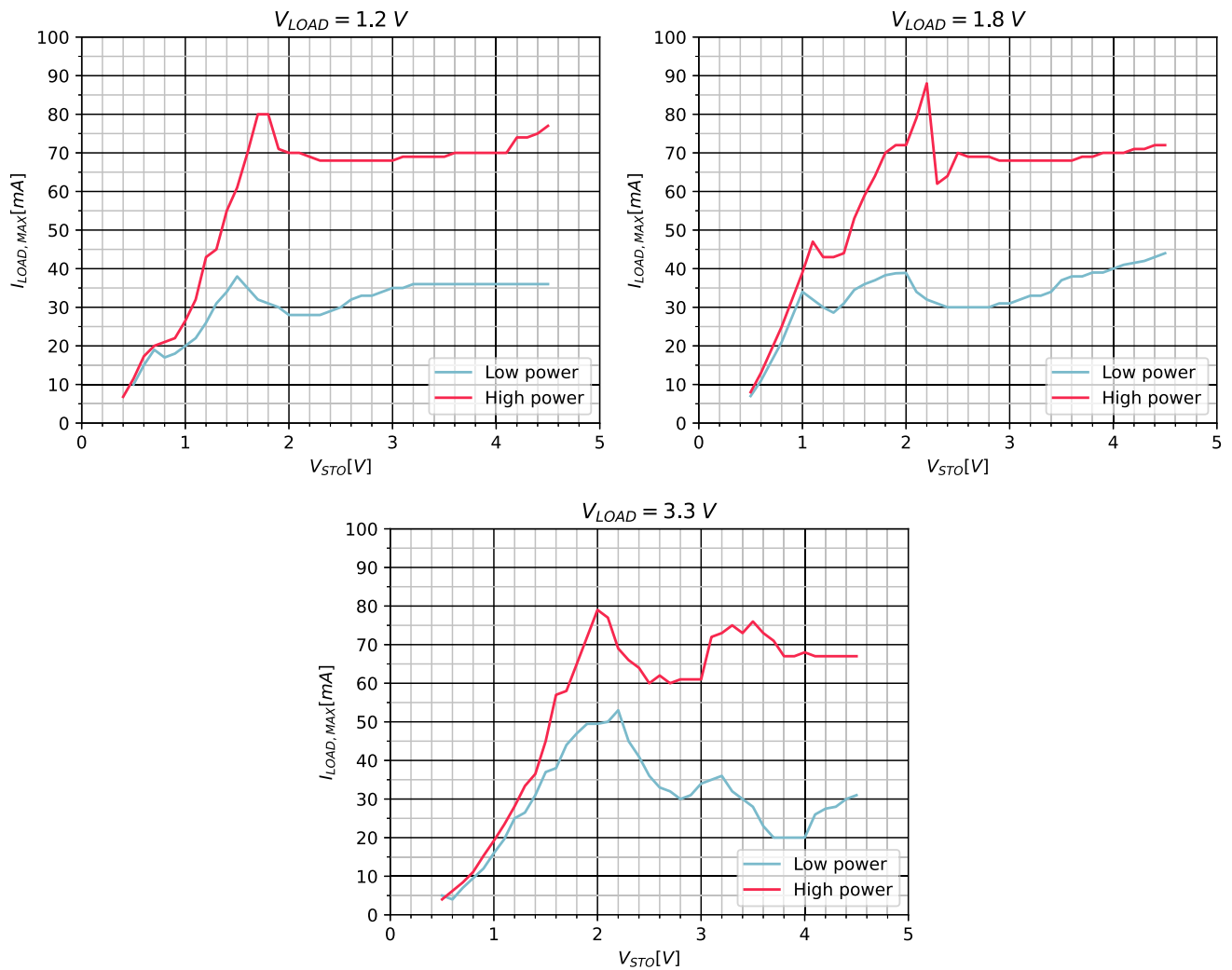


Figure 7: Maximum LOAD current depending on V_{STO} and V_{LOAD}

3.6.6. Quiescent Current

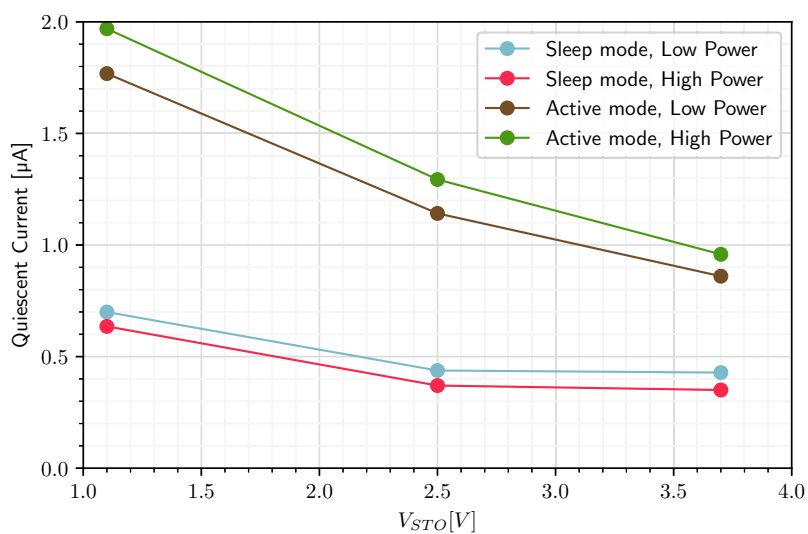


Figure 8: Quiescent current

4. Functional Block Diagram

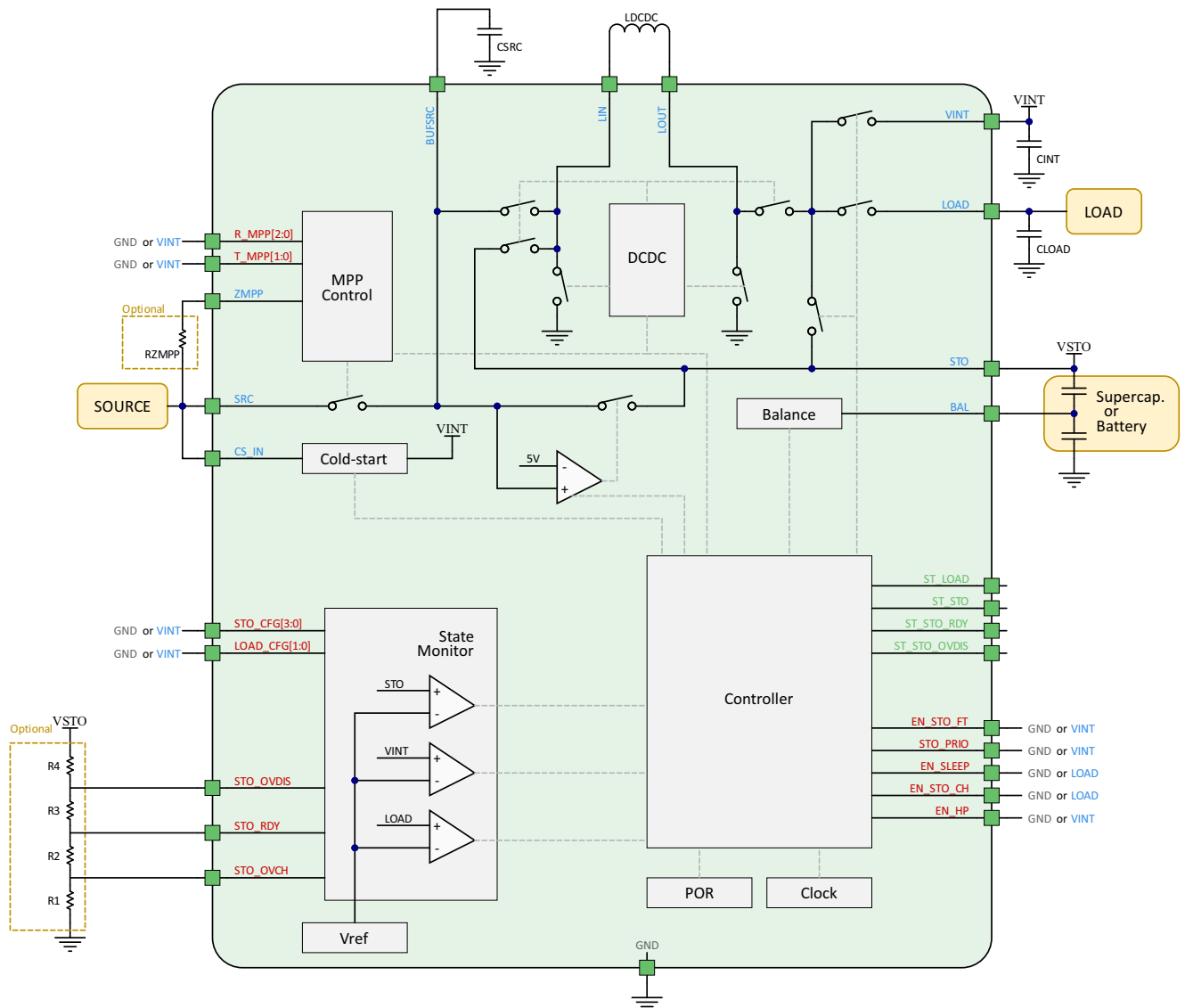


Figure 9: Functional Block Diagram

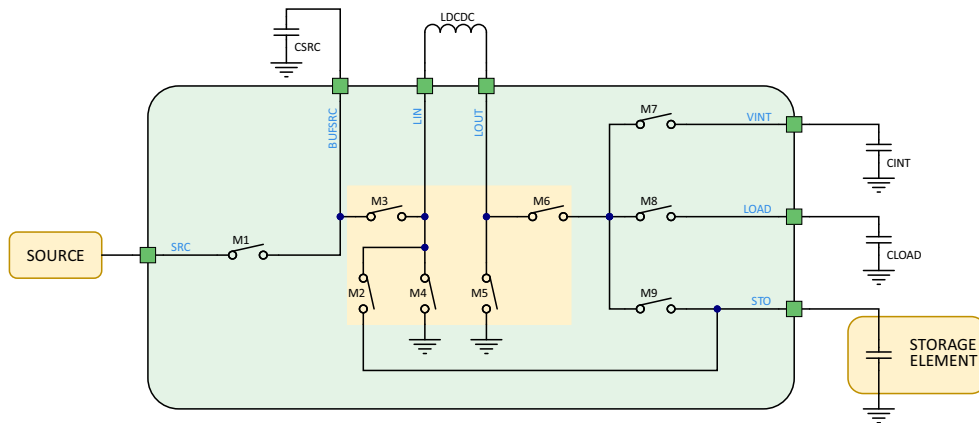


Figure 10: Simplified Schematic View of the AEM10330

5. Theory of Operation

5.1. DCDC Converter

The DCDC converter converts the voltage available at **BUFSRC** or at **STO** to a level suitable for charging the storage element **STO** or to regulate the **LOAD** and the internal supply **V_{VINT}**. The switching transistors of the DCDC converter are M2 or M3, M4, M5 and M6. Thanks to M7, M8 and M9, the controller selects between **LOAD**, **STO** and **VINT** respectively as the converter output. M1 selects the source as main input of energy. The internal supply **V_{VINT}** is regulated with priority over **LOAD**. **STO** is selected as an output only when neither **VINT** nor **LOAD** need to be supplied. The converter has two possible inputs: **BUFSRC** or **STO**. **BUFSRC** is used by default as an input via M3. If the energy available on **SRC** is not sufficient to maintain the **LOAD** or **VINT** voltage, for instance because of a sudden current peak on **LOAD**, the converter uses **STO** instead as an input via M2 to keep **LOAD** and **VINT** regulated.

The reactive power component of this converter is the external inductor **L_{DCDC}**. Periodically, the MPP control circuit disconnects the source from **BUFSRC** pin with the transistor M1 in order to let the harvester on **SRC** rise to its open-circuit voltage **V_{OC}** and measure it. This is done to define the optimal voltage level **V_{MPP}**, which is determined by applying the MPP ratio on **V_{OC}**. **BUFSRC** is decoupled by the capacitor **C_{SRC}**, which smooths the voltage against the current pulses pulled by the DCDC converter. The storage element is connected to the **STO** pin.

Depending on its input voltage and its output voltage, the DCDC converter will work as a boost converter, a buck converter or a buck-boost converter. The maximum power that can be harvested and supplied to the output **LOAD** depends on the power mode (**HIGH POWER MODE** or **LOW POWER MODE**), which is configured through the **EN_HP** pin (see Section 6.4).

DCDC Converter Mode	Input Voltage / Output Voltage
Boost	$V_{IN} < V_{OUT} - 250 \text{ mV}$
Buck	$V_{IN} > V_{OUT} + 250 \text{ mV}$
Buck - Boost	$V_{OUT} - 250 \text{ mV} < V_{IN} < V_{OUT} + 250 \text{ mV}$

Table 8: DCDC Converter Modes

5.2. Cold-Start Circuit

The AEM10330 is able to coldstart if the voltage on **CS_IN** is above 0.275 V. The minimum available power is:

- 3 μW if **V_{STO}** is above **V_{CHRDY}**.
- 6 μW if **V_{STO}** is below **V_{CHRDY}**.

CS_IN is typically connected to **SRC** to allow the AEM10330 to coldstart from the energy available on the harvester. Nevertheless, any other energy source can be connected to **CS_IN** as long as it meets the electrical specifications constraints described in Sections 3.4 and 3.5.

5.3. AEM10330 States Description

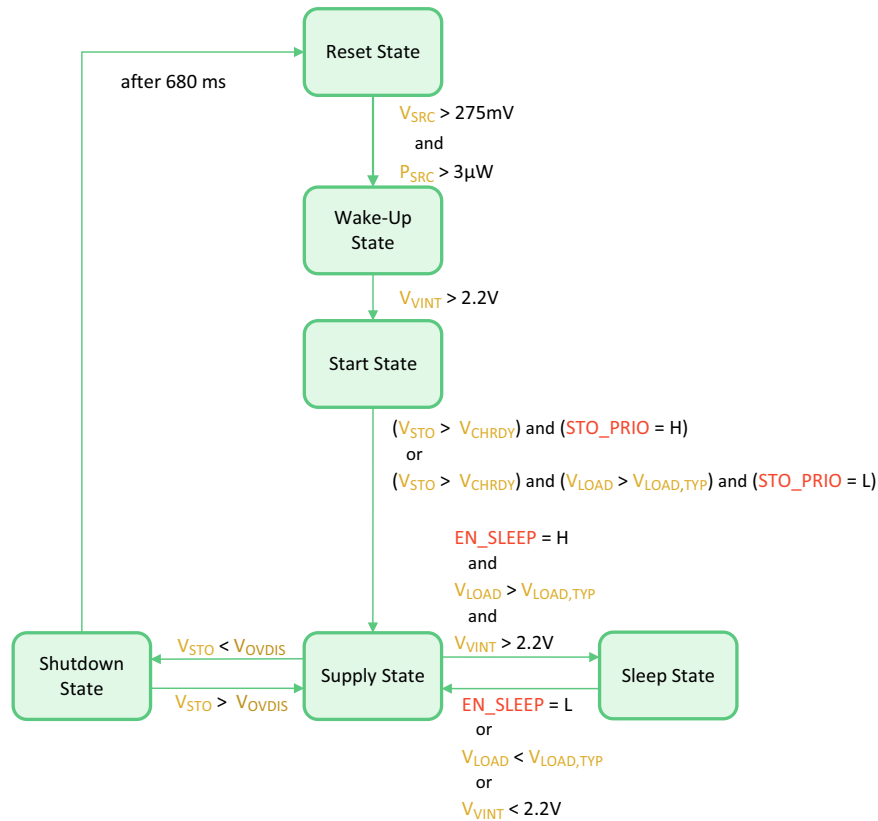


Figure 11: Diagram of the AEM10330 States

5.3.1. Reset, Wake Up and Start States

The **RESET STATE** is a state where all nodes are deeply discharged and there is no available energy to be harvested. As soon as the required cold start voltage of 275 mV and a sparse amount of power of just 3 μ W become available on **CS_IN** (usually connected to **SRC**), the AEM10330 switches to **WAKE-UP STATE**, and energy is extracted from **SRC** to make **V_VINT** rise to 2.2 V. When **V_VINT** reaches 2.2 V, the AEM10330 switches to **START STATE**. In **START STATE**, two scenarios are possible: in the first scenario, **STO_Prio** is HIGH, the storage element connected to **STO** has the priority on the one connected to **LOAD**. In the second scenario, **STO_Prio** is LOW and the **LOAD** has the priority.

When the AEM10330 is in **RESET STATE**, **WAKE-UP STATE** or **START STATE**, the DCDC converter's input is always **BUFSRC**. **STO** is never used as input. This guarantees that the storage element is not used until a minimum amount of energy has been stored in it.

5.3.1.1. Storage Element Priority

If **STO_Prio** is pulled up to **VINT**, the storage element connected on **STO** has priority over **LOAD**.

Supercapacitor as a Storage Element

If the storage element is a supercapacitor, it may be fully discharged at first and thus, need to be charged from 0 V. The DCDC converter charges **STO** from the input source (**SRC**). When **V_STO** reaches **V_CHRDY**, the circuit enters **SUPPLY STATE**.

Battery as a Storage Element

If the storage element is a battery, but its voltage is lower than **V_CHRDY**, then the storage element needs to be charged first until it reaches **V_CHRDY**. Once **V_STO** reaches **V_CHRDY**, or if the battery was initially charged above **V_CHRDY**, the circuit enters **SUPPLY STATE**.

5.3.1.2. Load Priority

If **STO_PRIO** is connected to GND, the AEM charges first the **LOAD** to $V_{LOAD,MAX}$ (see Table 10) using energy from the source (**SRC**). This allows to first supply the application circuit connected to **LOAD**. If the storage element was initially charged above V_{CHRDY} , the circuit enters **SUPPLY STATE** as soon as **LOAD** reaches $V_{LOAD,TYP}$. If the storage element is a supercapacitor or a battery which voltage is lower than V_{CHRDY} , the AEM keeps regulating **LOAD** between $V_{LOAD,MAX}$ and $V_{LOAD,TYP}$. Meanwhile, any excess charges on the source is used to charge the storage element until it reaches V_{CHRDY} . Once V_{STO} exceeds V_{CHRDY} , the circuit enters into **SUPPLY STATE**.

This configuration is useful when a large storage element is connected to **STO** and a smaller one is connected to **LOAD**: the application starts as soon as C_{LOAD} is charged and does not have to wait for the large storage element on **STO** to be charged.

5.3.2. Supply State

In **SUPPLY STATE**, four scenarios are possible:

- There is enough power provided by the source (**SRC**) to keep V_{VINT} at 2.2 V, V_{LOAD} near $V_{LOAD,TYP}$ with a small hysteresis and to charge the storage element. When the voltage on **VINT** falls below its 2.2 V regulation set point because of the AEM10330 internal circuit current consumption discharging C_{INT} , the DCDC converter switches its output to recharge C_{INT} from **SRC**, thus, keeping V_{VINT} regulated. The rest of the time, V_{LOAD} is maintained around $V_{LOAD,TYP}$ and the excess of power on source is used to charge the storage element on **STO**. In this case, the circuit remains in **SUPPLY STATE**. If **STO** is fully charged, **LOAD** will be maintained at $V_{LOAD,MAX}$ instead of $V_{LOAD,TYP}$. If all nodes are fully charged, the DCDC converter is disabled to prevent over-charging the storage element, and the **SRC** pin is set to high impedance.
- If the circuit connected to **LOAD** consumes more energy than the energy that the AEM10330 is able to extract from the source, the **LOAD** circuit will be supplied by the storage element connected on **STO**. In this case, the circuit stays in **SUPPLY STATE**.
- Due to a lack of power from the source, V_{STO} falls below V_{OVDIS} . In this case, the circuit enters **SHUTDOWN STATE** as explained in Section 5.3.3.
- If **EN_SLEEP** is HIGH and conditions (shown on Figure 11) on V_{LOAD} and V_{VINT} are satisfied, the AEM enters **SLEEP STATE** (see Section 5.3.4).

5.3.3. Shutdown State

If the storage element gets depleted ($V_{STO} < V_{OVDIS}$), the AEM10330 goes to **SHUTDOWN STATE**. As long as the AEM10330 is in this state, the **ST_STO_OVDIS** is HIGH. In **SHUTDOWN STATE**, if V_{STO} recovers within 680 ms, the AEM10330 goes back to **SUPPLY STATE**. This prevents false detection of the storage element being empty because of a **LOAD** current peak.

5.3.4. Sleep State

SLEEP STATE reduces the AEM10330 quiescent current by disabling the DCDC converter and by reducing the controller clock frequency. If V_{VINT} or V_{LOAD} fall below their regulation value, the AEM10330 temporarily exits **SLEEP STATE** to wake up the DCDC converter and supply **VINT** or **LOAD**. Exiting **SLEEP STATE** and waking up the DCDC converter takes up to 1 ms. Depending on the expected **LOAD** current, C_{LOAD} value must be adapted to act as an energy buffer during the 1 ms required to wake up the DCDC converter. Therefore, this state may be enabled when **LOAD** current is small, and be disabled if a high **LOAD** current is expected. As the DCDC is disabled, no energy is harvested from **SRC** while in **SLEEP STATE**, so that the storage element connected to **STO** is no longer charged. **LOAD** and **VINT** are the only nodes that are still supplied and regulated when the AEM10330 is in **SLEEP STATE**. This mode is useful to limit the current drawn on **STO** when the application circuit has detected that the power available on **SRC** is so low that it wouldn't compensate the AEM10330 internal circuit power consumption (see Table 6).

The AEM10330 enters **SLEEP STATE** if all the following conditions are satisfied:

- **EN_SLEEP** pin pulled up to **LOAD** (H)
- $V_{VINT} > 2.2 \text{ V}$
- $V_{LOAD} > V_{LOAD,TYP}$

The AEM10330 leaves **SLEEP STATE** and switches back to **SUPPLY STATE** if one of the following conditions is satisfied:

- **EN_SLEEP** pin pulled down to GND (L)
- $V_{VINT} < 2.2 \text{ V}$
- $V_{LOAD} < V_{LOAD,TYP}$

The AEM10330 will then stay in **SUPPLY STATE** until the **SLEEP STATE** conditions are all satisfied again.

5.4. Maximum Power Point Tracking

In **SUPPLY STATE**, **SHUTDOWN STATE** and **START STATE**, the voltage on **SRC** is regulated by an internal Maximum Power Point Tracking (MPPT) module. The MPPT module evaluates V_{MPP} , the voltage at which the source provides the highest possible power, as a given fraction of the open-circuit voltage of the source (V_{OC}). This ratio is set by the **R_MPP[2:0]**

terminals according to Table 11. The AEM10330 supports any V_{MPP} levels in the range from 100 mV to 4.5 V. It offers a choice of seven values for the V_{MPP} / V_{OC} fraction. It can also match the input impedance of the DCDC converter with an impedance connected to the Z_{MPP} terminal as explained in Section 6.7.

The MPPT module evaluates periodically the open-circuit voltage (V_{OC}) with the following sequence to ensure optimal power extraction at any time:

- The AEM10330 stops extracting power from the SRC during $T_{MPPT, WAIT}$ to allow the SRC voltage to rise to V_{OC} .
- Once this delay elapses, the AEM10330 performs the measurement of V_{OC} during $T_{MPPT, MEASURE}$.
- After the measurement, the AEM10330 resumes extracting power by regulating the SRC voltage to the newly determined V_{MPP} .
- This MPP evaluation is repeated every $T_{MPPT, PERIOD}$.

$T_{MPPT, WAIT}$ and $T_{MPPT, PERIOD}$ are set according to Table 12 by the $T_{MPP}[1:0]$ terminals while $T_{MPPT, MEASURE}$ is constant for any configuration (see Table 6).

The total time during which the AEM10330 does not extract power correspond to the sum of $T_{MPPT, WAIT}$ and $T_{MPPT, MEASURE}$ (see Figure 12).

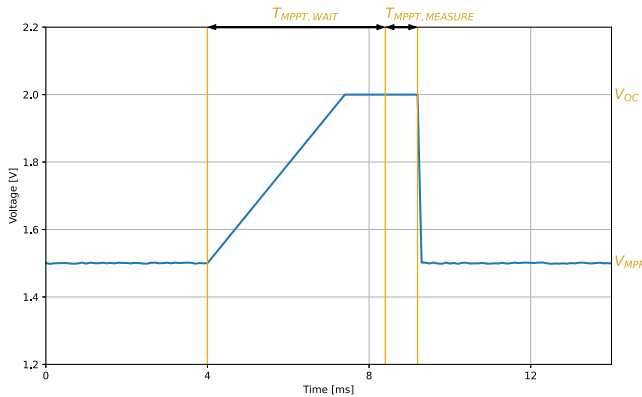


Figure 12: MPPT evaluation behavior (with $T_{MPPT, WAIT}$ of 4.4 ms)

5.5. Balancing for Dual-Cell Supercapacitor

The AEM10330 balancing feature allows to balance the internal voltage of a dual-cell supercapacitor connected to STO in order to avoid damaging the supercapacitor because of excessive voltage on one cell.

If a battery, a capacitor or a single-cell supercapacitor is connected on STO , the balancing feature is not needed. In this case, BAL must be connected to GND to disable the balancing.

If a dual-cell supercapacitor is connected on STO , BAL must be connected to the node between both cells of the supercapacitor. The balancing circuit will compensate for any mismatch of the two cells that could lead to the overcharge of one of two cells. The balancing circuit ensures that BAL remains close to $V_{STO} / 2$.

The balancing circuit works as follows:

- The balancing circuit is enabled only when $V_{BAL} > V_{BAL, MIN}$ (see Table 6).
- If $V_{BAL} > \frac{V_{STO}}{2}$: the AEM10330 enables a switch between BAL and GND to discharge the bottom supercapacitor cell to GND (up to 20 mA).
- If $V_{BAL} < \frac{V_{STO}}{2}$: the AEM10330 enables a switch between STO and BAL to discharge the top supercapacitor cell to the bottom supercapacitor cell (up to 20 mA).

6. System Configuration

6.1. Storage Element Configuration

Through four configuration pins (**STO_CFG[3:0]**), the user can set a particular operating mode from a range that covers most application requirements, without any dedicated external component as shown in Table 9. The three threshold levels are defined as:

- **V_{OVCH}**: maximum voltage accepted on the storage element before disabling its charging.
- **V_{CHRDY}**: minimum voltage required on the storage element before starting to supply the **LOAD** (if **STO_PRIO** is HIGH), entering supply state after start-up and setting **ST_STO** HIGH.
- **V_{OVDIS}**: minimum voltage accepted on the storage element before considering the storage element as depleted, thus, setting **ST_STO** LOW.

A large-size storage element is not mandatory on **STO**:

- If the harvested energy source is permanently available and covers the application needs or
- If the application does not need to store energy when the harvested energy source is not available

The storage element may then be replaced by an external capacitor **C_{STO}** with a minimum value of 100 μ F.

CAUTION: running the AEM10330 without this 100 μ F minimum capacitance on **STO** will permanently damage the circuit.

DISCLAIMER: storage element thresholds provided in the table below are indicative to support a wide range of storage element variants. They are provided as is to the best knowledge of e-peas's application laboratory. They should not replace the actual values provided in the storage element manufacturer's specifications and datasheet.

Configuration pins				Storage element threshold voltages			Typical use
STO_CFG[3:0]				V_{OVDIS}	V_{CHRDY}	V_{OVCH}	
L	L	L	L	3.00 V	3.50 V	4.05 V	LiCoO ₂ battery, Li-Po battery, Lithium Titanate (3.8 V) battery (long life).
L	L	L	H	2.80 V	3.10 V	3.60 V	LiFePO ₄ battery, Lithium capacitor (LiC).
L	L	H	L	1.85 V	2.40 V	2.70 V	Dual-cell NiMH battery, Lithium-Titanate (2.4V) battery.
L	L	H	H	0.20 V	1.00 V	4.65 V	Dual-cell supercapacitor.
L	H	L	L	0.20 V	1.00 V	2.60 V	Single-cell supercapacitor.
L	H	L	H	1.00 V	1.20 V	2.95 V	Single-cell supercapacitor.
L	H	H	L	1.85 V	2.30 V	2.60 V	Lithium-Titanate battery (2.4V).
L	H	H	H	Custom Mode (single-cell NiMH battery, LiC, etc.) ¹ .			
H	L	L	L	1.10 V	1.25 V	1.50 V	Ni-Cd single-cell battery.
H	L	L	H	2.20 V	2.50 V	3.00 V	Ni-Cd dual-cell battery.
H	L	H	L	1.45 V	2.00 V	4.65 V	Dual-cell supercapacitor.
H	L	H	H	1.00 V	1.20 V	2.60 V	Single-cell supercapacitor.
H	H	L	L	2.00 V	2.30 V	2.60 V	Solid State battery.
H	H	L	H	3.00 V	3.50 V	4.35 V	LiCoO ₂ battery, Li-Po battery, Lithium Titanate (3.8 V) battery.
H	H	H	L	2.60 V	2.70 V	4.00 V	Tadiran TLI.
H	H	H	H	2.60 V	3.50 V	3.90 V	Tadiran HLC.

Table 9: Storage Element Configuration Pins

1. An example of a single-cell NiMH batteries optimized custom mode setting can be found at Section 7.2.

6.2. Custom Mode Configuration

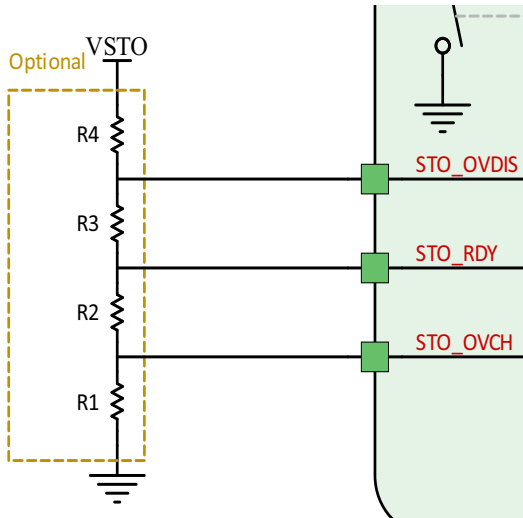


Figure 13: Custom Mode Settings

When **STO_CFG[3:0] = LHHH**, the custom mode is selected and all four configuration resistors must be wired as shown in Figure 13. V_{OVCH} , V_{CHRDY} and, V_{OVDIS} are defined thanks to R1, R2, R3 and R4, which can be determined within the following constraints:

- $R_T = R_1 + R_2 + R_3 + R_4$
- $1\text{M}\Omega \leq R_T \leq 100\text{M}\Omega$
- $R_1 = R_T \cdot \frac{1\text{V}}{V_{OVCH}}$
- $R_2 = R_T \cdot \left(\frac{1\text{V}}{V_{CHRDY}} - \frac{1\text{V}}{V_{OVCH}} \right)$
- $R_3 = R_T \cdot \left(\frac{1\text{V}}{V_{OVDIS}} - \frac{1\text{V}}{V_{CHRDY}} \right)$
- $R_4 = R_T \cdot \left(1 - \frac{1\text{V}}{V_{OVDIS}} \right)$

The resistors should have high values to make the additional power consumption negligible. Moreover, the following constraints must be respected to ensure the functionality of the chip:

- $V_{CHRDY} + 0.05\text{V} \leq V_{OVCH} \leq 4.5\text{V}$
- $V_{OVDIS} + 0.05\text{V} \leq V_{CHRDY} \leq V_{OVCH} - 0.05\text{V}$
- $1\text{V} \leq V_{OVDIS}$

6.3. Disable Storage Element Charging

Pulling down **EN_STO_CH** pin to GND disables the charging of the storage element connected to **STO** from **SRC**. This can be done for example to protect the storage element when the system detects that the environment temperature is too low or too high to safely charge the storage element.

While **EN_STO_CH** is pulled down, **VINT** and **LOAD** can still be supplied either from **SRC** or from **STO**.

To enable charging the storage element on **STO**, **EN_STO_CH** must be pulled up to **LOAD** or left floating (pin is pulled up internally).

Please note that **STO** will still be charged to V_{CHRDY} during the **START STATE**.

6.4. High Power / Low Power Mode

When **EN_HP** is pulled to **VINT**, the DCDC converter is configured to **HIGH POWER MODE**. This allows higher currents to be extracted from the DCDC converter input (**SRC** or **STO**) to the DCDC converter output (**LOAD** or **STO**) with a trade-off in efficiency. Figure 7 shows the maximum current that the DCDC converter can supply to **LOAD**, depending on the storage voltage V_{STO} , for every available load voltage V_{LOAD} , for both **HIGH POWER MODE** and **LOW POWER MODE**.



6.5. Load Configuration

The **LOAD** output voltage V_{LOAD} can be configured thanks to the **LOAD_CFG[1:0]** configuration pins to cover most application cases (see Table 10).

When the AEM10330 is in **SUPPLY STATE**:

- $V_{LOAD,TYP}$ is the typical average voltage to be expected on the **LOAD** pin.
- When $V_{LOAD} < V_{LOAD,MID}$: the AEM10330 charges C_{LOAD} from the storage element (**STO**) if V_{STO} is above V_{OVDIS} , even if energy is available on **SRC**.
- $V_{LOAD,MID} < V_{LOAD} < V_{LOAD,MAX}$:
 - If energy is available on **SRC**, the AEM10330 uses it to charge C_{LOAD} .
 - If no energy is available on **SRC**, the AEM10330 uses the energy available on **STO** to charge C_{LOAD} .

- **ST_LOAD** reflects V_{LOAD} status:
 - **ST_LOAD** is HIGH when $V_{LOAD} > V_{LOAD,TYP}$.
 - **ST_LOAD** is LOW when $V_{LOAD} < V_{LOAD,MIN}$.
- As a reminder, the order of priority in which the nodes are recharged is as follows:
 - **VINT** always has the highest priority, as it is mandatory to keep the AEM10330 internal circuit supplied.
 - **LOAD** has the second highest priority, to keep the application circuit supplied.
 - **STO** has the lowest priority, as the storage element is used to store the excessive energy from **SRC**.

Configuration pins		LOAD output voltage			
LOAD_CFG[1:0]		$V_{LOAD,MIN}$	$V_{LOAD,MID}$	$V_{LOAD,TYP}$	$V_{LOAD,MAX}$
L	L	3.15 V	3.23 V	3.28 V	3.34 V
L	H	2.35 V	2.47 V	2.50 V	2.53 V
H	L	1.64 V	1.75 V	1.79 V	1.82 V
H	H	1.14 V	1.16 V	1.20 V	1.23 V

Table 10: Load Configuration Pins

6.6. MPPT Configuration

The MPPT module is configured through the following pins:

- **R_MPP[2:0]** allows for configuring the V_{MPP} / V_{OC} tracking ratio, that has to be chosen according to the characteristics of the source. Please note that if the selected mode is **ZMPP**, an external resistor must be added to the circuit as explained in Section 6.7.
- **T_MPP[1:0]** allows for configuring the duration of an MPP evaluation and the time between two MPP evaluations.

Configuration pins			MPPT ratio
R_MPP[2:0]			V_{MPP} / V_{OC}
L	L	L	60%
L	L	H	65%
L	H	L	70%
L	H	H	75%
H	L	L	80%
H	L	H	85%
H	H	L	90%
H	H	H	ZMPP

Table 11: MPPT Ratio Configuration Pins

Configuration pins		MPPT timing	
T_MPP[1:0]		$T_{MPPT, WAIT} + T_{MPPT, MEASURE}^1$	$T_{MPPT, PERIOD}$
L	L	5.19 ms	280 ms
L	H	70.8 ms	4.5 s
H	L	280 ms	17.9 s
H	H	1.12 s	71.7 s

Table 12: MPPT Timing Configuration Pins

1. The total duration of the open-circuit voltage evaluation is the sum of $T_{MPPT, WAIT}$ and $T_{MPPT, MEASURE}$. $T_{MPPT, MEASURE}$ is fixed, see Table 6).

6.7. ZMPP Configuration

Instead of working at a ratio of the open-circuit voltage, the AEM10330 can regulate the input impedance of the DCDC converter so that it matches a constant impedance R_{ZMPP} connected to the **ZMPP** pin (see Figure 14). In this case, the AEM10330 regulates V_{SRC} at a voltage that is the product of the **ZMPP** resistance R_{ZMPP} and the current available at the **SRC** input.

$$10 \Omega \leq R_{ZMPP} \leq 100 \text{ k}\Omega$$

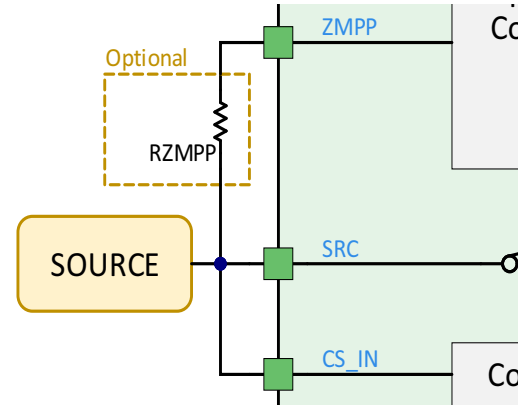


Figure 14: R_{ZMPP} Connection to the AEM10330

6.8. Source to Storage Element Feed-Through

When the harvester connected to **SRC** delivers a high amount of power, the AEM might not be able to pull enough current to regulate V_{SRC} to the MPP voltage, thus, increasing the voltage on **SRC**, eventually above 5V. To maximize the energy extracted in that case, the AEM10330 can be configured to create a direct feed-through current path from **SRC** to **STO** when V_{SRC} is above 5V. This is measured when the AEM is pulling current from the source (not during an MPP evaluation).

If the MPPT module detects that V_{SRC} is higher than 4 V and **EN_STO_FT** is set HIGH, the **SRC** is monitored. From that moment, if the AEM10330 detects that V_{SRC} rises above 5 V and if the storage element is not fully charged, the switch between **SRC** and **STO** pins is closed until V_{SRC} drops below 5 V or until the storage element is fully charged.

This feature is enabled by pulling up **EN_STO_FT** pin to **VINT**. However, it is disabled if the storage element is fully charged, or when a MPP evaluation is occurring. Therefore the circuit must still be protected from any overshoot voltage on **SRC** above 5.5 V, for instance by a zener diode.

6.9. External Components

Refer to Figure 19 to have an illustration of the external components wiring.

6.9.1. Storage Element Information

The energy storage element of the AEM10330 can be a rechargeable battery, a supercapacitor or a capacitor. The size of the storage element must be determined so that its voltage does not fall below V_{OVDIS} even during current peaks pulled by the application circuit connected to **LOAD**. If the internal resistance of the storage element cannot sustain this voltage limit, it is advisable to decouple the battery with a capacitor.

If the application expects a disconnection of the battery (e.g. because of a user removable connector), the PCB must include a capacitor C_{STO} of at least 100 μF connected between **STO** and **GND**. The leakage current of the storage element should be small as leakage currents directly impact the quiescent current of the whole subsystem.

6.9.2. External Inductor Information

The AEM10330 operates with one standard miniature inductor. L_{DCDC} must sustain a peak current of at least 1 A and a switching frequency of at least 10 MHz. Low equivalent series resistance (ESR) favors the power conversion efficiency of the DCDC converter. The recommended value is 10 μH .

6.9.3. External Capacitors Information

6.9.3.1. C_{SRC}

This capacitor acts as an energy buffer at the input of the DCDC converter. It prevents large voltage fluctuations when the DCDC converter is switching. The recommended nominal value is 22 μF .

6.9.3.2. C_{INT}

This capacitor acts as an energy buffer for the internal voltage supply. The recommended nominal value is 10 μF .

6.9.3.3. C_{LOAD}

This capacitor acts as an energy buffer for **LOAD**. It also reduces the voltage ripple induced by the current pulses inherent to the switched behavior of the converter. The recommended value is at least 13 μF (considering derating and tolerance).

7. Typical Application Circuits

7.1. Example Circuit 1

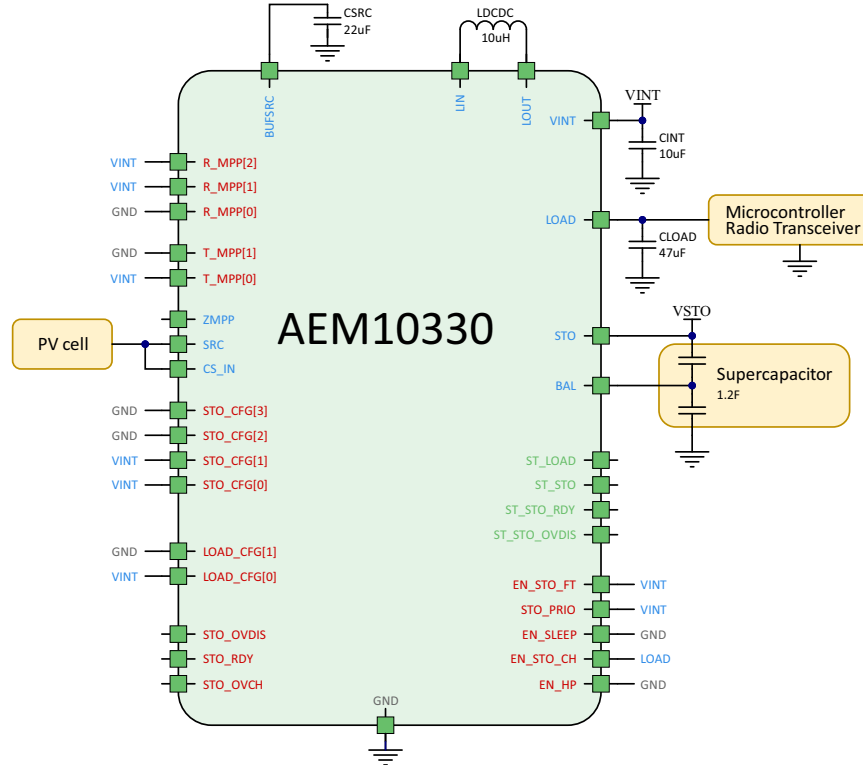


Figure 15: Typical Application Circuit 1

The circuit is an example of a system with solar energy harvesting. It uses a pre-defined operating mode that uses standard components, and a supercapacitor as energy storage.

- Energy source: PV cell.
- **R_MPP[2:0]** = HHL: the MPP tracker ratio is set to 90%.
- **T_MPP[1:0]** = LH: the MPP evaluation period is 4.5 s and the MPP wait duration is 70.8 ms.
- **STO_CFG[3:0]** = LLHH: the storage element is a dual-cell supercapacitor, with:
 - $V_{OVCH} = 4.65 \text{ V}$
 - $V_{CHRDY} = 1.00 \text{ V}$
 - $V_{OVDIS} = 0.20 \text{ V}$
- The balancing pin of the dual-cell supercapacitor is connected to **BAL**.
- **LOAD_CFG[1:0]** = LH: the micro-controller and the radio transceiver are supplied by the **LOAD** terminal, which is regulated at $V_{LOAD} = 2.50 \text{ V}$.
- **STO_Prio** is connected to **VINT**: at start-up **STO** will be charged before **LOAD**.
- **EN_SLEEP** is connected to **GND**: the AEM10330 will never switch to **SLEEP STATE**.
- **EN_STO_CH** is connected to **LOAD**: the charging of the storage element on **STO** is enabled.
- **EN_HP** is connected to **GND**: the DCDC converter is in **LOW POWER MODE**.
- **EN_STO_FT** is connected to **VINT**: **SRC** to **STO** feed-through is enabled when V_{SRC} is above 5 V.

7.2. Example Circuit 2

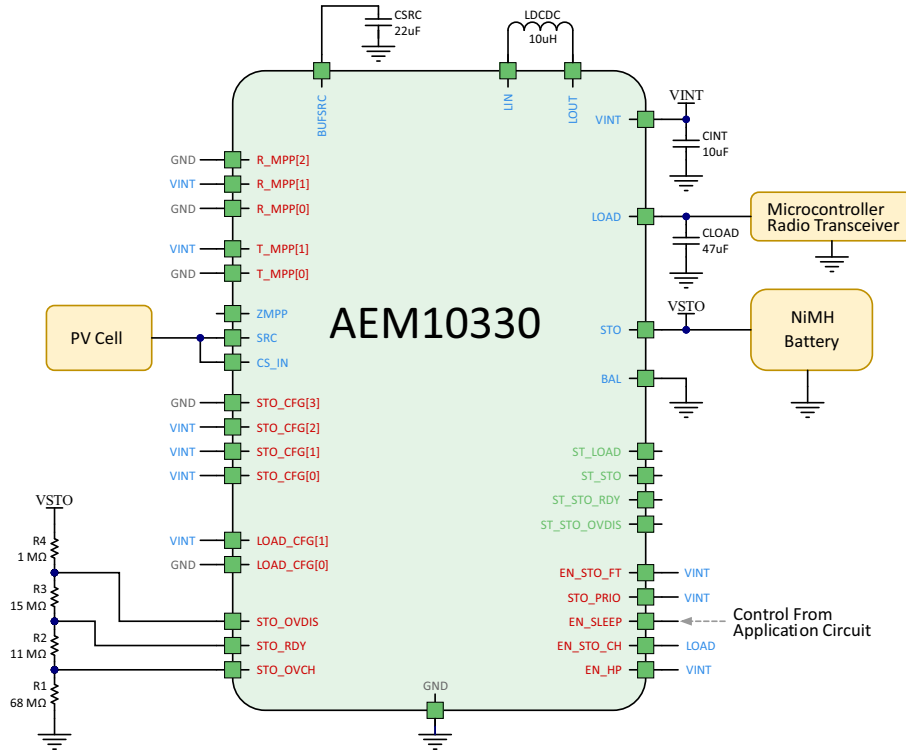


Figure 16: Typical Application Circuit 2

The circuit is an example of a system with solar energy harvesting. It uses a rechargeable NiMH battery as storage element. The voltage thresholds are set by the custom mode.

- Energy source: PV cell.
- $R_MPP[2:0]$ = LHL: the MPP ratio is set to 70%.
- $T_MPP[1:0]$ = HL: the MPP evaluation period is 17.9 s and the MPP wait duration is 280 ms.
- $STO_CFG[3:0]$ = LHHH: the storage element is a NiMH rechargeable battery, used with custom mode:
 - $V_{OVDIS} = 1.00\text{ V}$
 - $V_{CHRDY} = 1.20\text{ V}$
 - $V_{OVCH} = 1.40\text{ V}$
- Custom mode resistor divider calculations (values have been chosen to match E24 series value):
 - $R_T = 95\text{ M}\Omega$
 - $R_1 = R_T \cdot \frac{1\text{ V}}{V_{OVCH}} \approx 68\text{ M}\Omega$
 - $R_2 = R_T \cdot \left(\frac{1\text{ V}}{V_{CHRDY}} - \frac{1\text{ V}}{V_{OVCH}} \right) \approx 11\text{ M}\Omega$
 - $R_3 = R_T \cdot \left(\frac{1\text{ V}}{V_{OVDIS}} - \frac{1\text{ V}}{V_{CHRDY}} \right) \approx 15\text{ M}\Omega$
 - $R_4 = R_T \cdot \left(1 - \frac{1\text{ V}}{V_{OVDIS}} \right) \approx 1\text{ M}\Omega$

- BAL is not used (not a dual-cell storage element) so it is connected to GND .
- $LOAD_CFG[1:0]$ = HL: the micro-controller and the radio transceiver are supplied by the $LOAD$ terminal, which is regulated at $V_{LOAD} = 1.79\text{ V}$.
- STO_PRIO is connected to $VINT$: at start-up STO will be charged before $LOAD$.
- EN_SLEEP is controlled by the application circuit, typically by a micro-controller GPIO output.
- EN_STO_CH is connected to $LOAD$: the charging of the storage element present on STO is enabled.
- EN_HP is connected to $VINT$: the DCDC converter is in **HIGH POWER MODE**.
- EN_STO_FT is connected to $VINT$: SRC to STO feed-through is enabled when V_{SRC} is above 5 V.

NOTE: for LiC (Lithium-ion Capacitor) storage elements, or others that would not be covered by $STO_CFG[3:0]$ presets, please apply the same equations as in the above example to determine custom mode resistors values. E24 series values for typical storage elements can be found in the AEM10330 Configuration Tool spreadsheet, to be downloaded on e-peas website.

8. Circuit Behavior

8.1. Wake-up state, Start state and Supply state

8.1.1. Configuration

- **STO_CFG[3:0]** = LLLL
 - V_{OVDIS} = 3.00 V
 - V_{CHRDY} = 3.50 V
 - V_{OVCH} = 4.05 V
- **R_MPP[2:0]** = LLL (60%)
- **T_MPP[1:0]** = LL (5.19 ms / 280 ms)
- C_{STO} = 10 mF
- **SRC**: 5 mA current source with 3 V voltage compliance
- **LOAD_CFG[1:0]** = LL ($V_{LOAD,TYP}$ = 3.28 V)
- **EN_HP** = H (high power mode)
- **EN_STO_CH** = H (storage element charge enabled)
- **STO_Prio** = H (storage element charged in priority after startup)
- **EN_SLEEP** = L (sleep mode disabled)

8.1.2. Observations

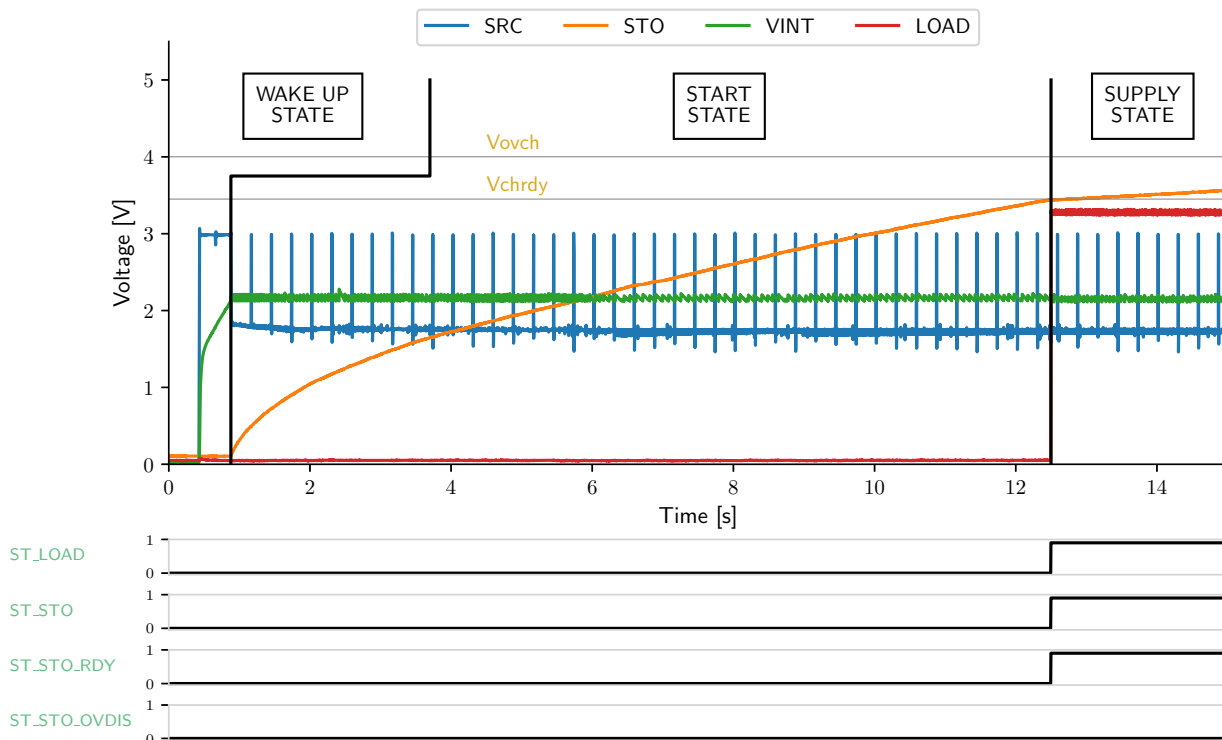


Figure 17: Wake-up state, Start state and Supply state

- The AEM10330 is initially in **WAKE-UP STATE**. All the status pins are LOW, the storage element connected on **STO** is depleted and no energy is applied on **SRC**.
- Once the supply connected on **SRC** is switched on, the AEM10330 coldstarts. V_{VINT} rises around 2.2 V, the AEM10330 switches to **START STATE**.
- In **START STATE**, the AEM10330 charges the storage element connected on **STO**. V_{SRC} is regulated to V_{MPP} thanks to the MPPT module. The open circuit voltage V_{OC} is measured every $T_{MPPT,PERIOD}$.
- Once V_{STO} reaches V_{CHRDY} , the AEM10330 switches to **SUPPLY STATE**.
- In **SUPPLY STATE**, the **LOAD** output is enabled, so that V_{LOAD} is regulated at 3.28 V. **ST_LOAD**, **ST_STO** and **ST_STO_RDY** pins are asserted. The AEM10330 keeps charging the storage element.

8.2. Supply state, Shutdown state and Reset state

8.2.1. Configuration

- **STO_CFG[3:0]** = LLLL
 - V_{OVDIS} = 3.00 V
 - V_{CHRDY} = 3.50 V
 - V_{OVCH} = 4.05 V
- **LOAD_CFG[1:0]** = LL ($V_{LOAD,TYP}$ = 3.28 V)
- **R_MPP[2:0]** = LLL (60%)
- **T_MPP[1:0]** = LL (5.19 ms / 280 ms)
- C_{STO} = 10 mF
- **SRC**: left floating to force the storage element on **STO** to discharge
- **EN_HP** = H (high power mode)
- **STO_PRIO** = H (storage element charged in priority after startup)
- **EN_STO_CH** = H (storage element charge enabled)
- **EN_SLEEP** = L (sleep mode disabled)
- 10 k Ω connected between **LOAD** and **GND**

8.2.2. Observations

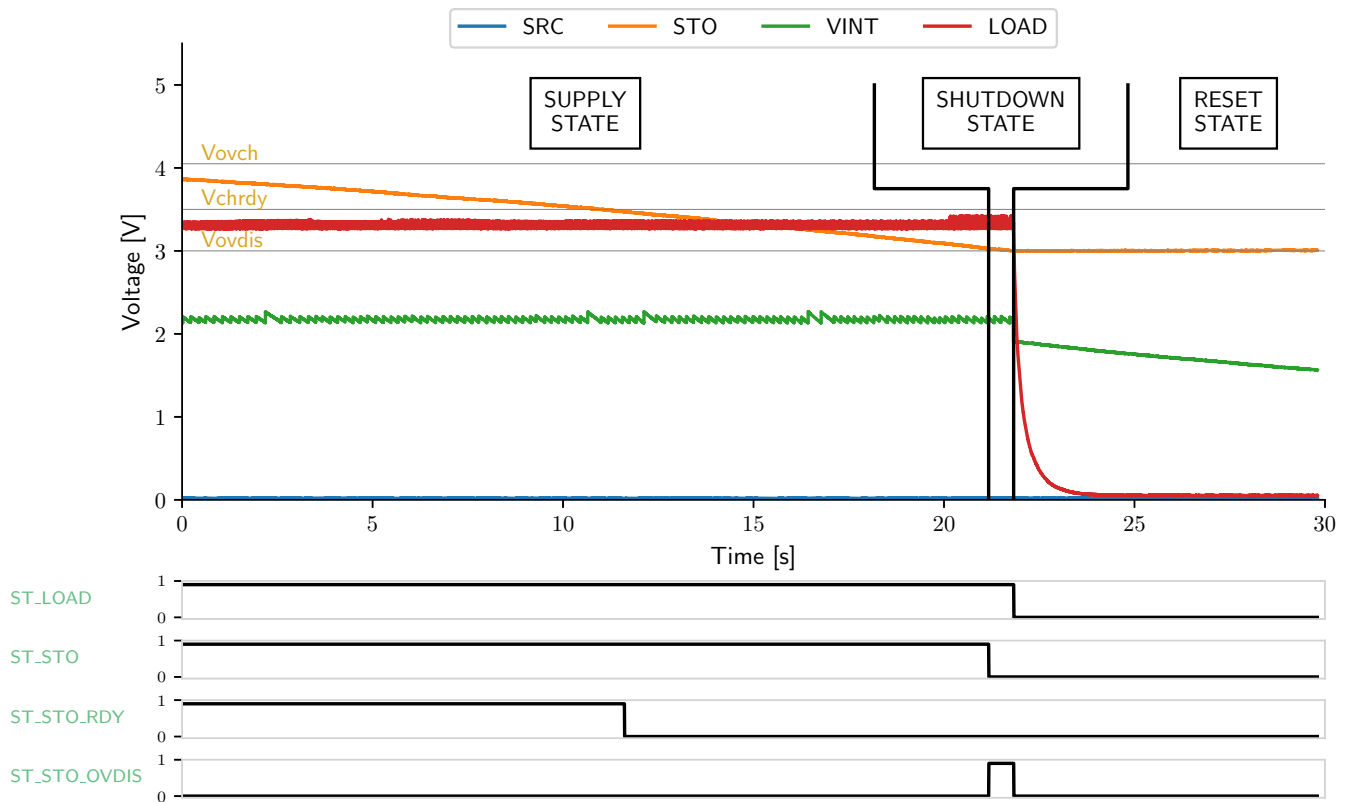


Figure 18: Supply state, Shutdown state and Reset state

- The AEM10330 is in **SUPPLY STATE** when the **SRC** pin is disconnected from the harvester and left floating. Causing the storage element connected on **STO** to discharge.
- Once V_{STO} discharge below V_{CHRDY} , **ST_STO_RDY** is set LOW. V_{LOAD} keeps being regulated.
- Once V_{STO} reaches V_{OVDIS} , the AEM10330 goes into **SHUTDOWN STATE**. **ST_STO_OVDIS** is HIGH as long as the AEM10330 stays in this state, and **ST_STO** is LOW.
- Since V_{STO} has not recovered above V_{OVDIS} within 680 ms, the AEM10330 goes into **RESET STATE**.
- In **RESET STATE**, **ST_LOAD** goes LOW. V_{LOAD} and V_{VINT} stop being regulated.



9. Schematic

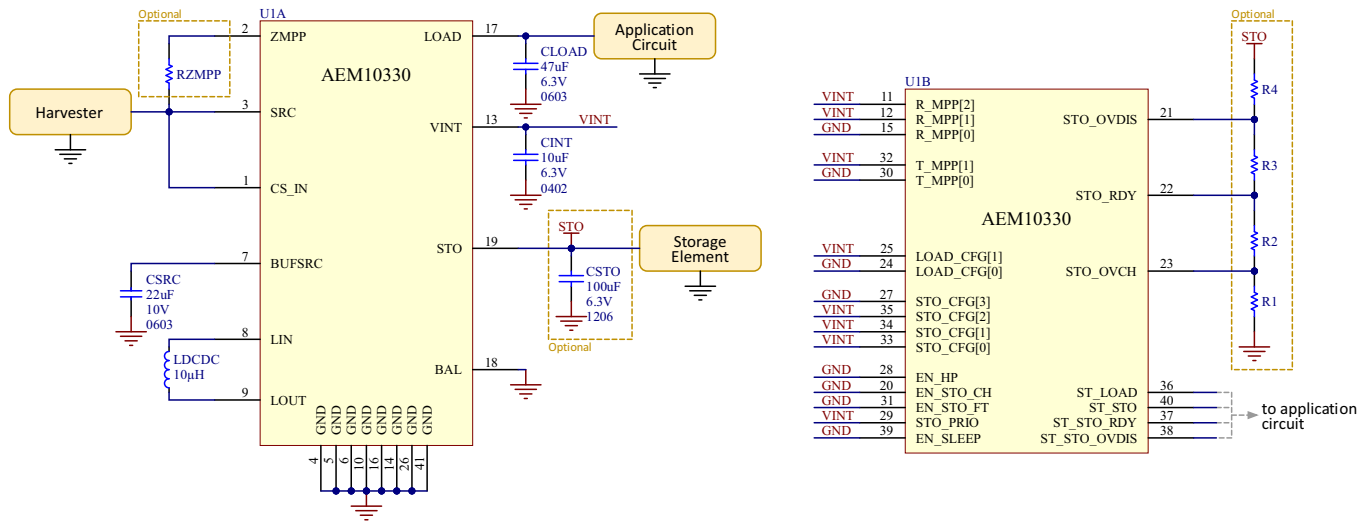


Figure 19: Schematic Example

Designator	Description	Quantity	Manufacturer	Link
U1	AEM10330 - Symbol QFN 40-pin	1	e-peas	order at sales@e-peas.com
L _{DCDC}	Power inductor 10 µH - 1.76A	1	Murata	DFE252010F-100M
C _{INT}	Ceramic Cap 10 µF, 6.3V, 20%, X5R 0402	1	Murata	GRM155R60J106ME15
C _{SRC}	Ceramic Cap 22 µF, 10V, 20%, X5R 0603	1	Murata	GRM188R61A226ME15
C _{LOAD}	Ceramic Cap 47 µF, 6.3V, 20%, X5R 0603	1	Murata	GRM188R60J476ME15
C _{STO} (optional)	Ceramic Cap 220 µF, 6.3V, 20%, X5R 1206	1	Murata	GRM31CR60J227ME11

Table 13: Minimal Bill of Materials

10. Layout

10.1. Guidelines

Good layout practices are mandatory in order to obtain good stability and best efficiency with the AEM10330. It also allows for minimizing electromagnetic interferences generated by the AEM10330 DCDC converter.

The following list, while not exhaustive, shows the main attention points when routing a PCB with the AEM10330:

- The switching nodes (LIN and LOUT) must be kept as short as possible, with minimal track resistance and minimal track capacitance. Low resistance is obtained by keeping track length as short as possible and track width as large as possible between L_{DCDC} and the AEM10330 pins. Minimal capacitance is obtained by keeping distance between LIN/LOUT and other signals. We recommend removing the ground plane, the power plane and the bottom layer ground pour under L_{DCDC} footprint, as well as adding distance between LIN/LOUT and the top ground pour, as shown on Figure 20.
- The DCDC decoupling capacitors (C_{SRC} - C_{LOAD} - C_{STO}) must be placed as close as possible to the AEM10330, with direct connection and minimum track resistance for the corresponding power nodes (BUFSRC, LOAD and STO).
- The GND return path between the DCDC decoupling capacitors (C_{SRC} - C_{LOAD} - C_{STO}) and the AEM10330 thermal pad, which is the AEM10330 main GND connection, must be as direct and short as possible. This is preferably done on the top layer when possible, otherwise by internal/bottom plane, using low resistance vias to decrease layer-to-layer connection resistance.
- The external DC power connections (SRC, LOAD and STO) must be connected to the AEM10330 with low resistance tracks.
- Connection between VINT and C_{INT} must be moderately short for AEM10330 stability, even though this pins does not carry large currents. Same for connection between C_{INT} to GND.
- If used, ZMPP must be connected to the AEM10330 with a low resistance track, according to the expected SRC power.
- The BAL pin connection track must be able to handle at least 40 mA.
- The custom mode setting pins STO_OVDIS, STO_RDY and STO_OVCH are high impedance analog inputs typically connected to a resistive divider with high resistor values, making those three nodes prone to pickup noise. Thus, it is recommended to keep those as short as possible and as far as possible to noise sources such as DCDC switching nodes.
- The configuration pins and the status pins have minimal layout restrictions. CS_IN maximum current is below 1 mA, so its layout restrictions are minimal as well.

10.2. Layout Example

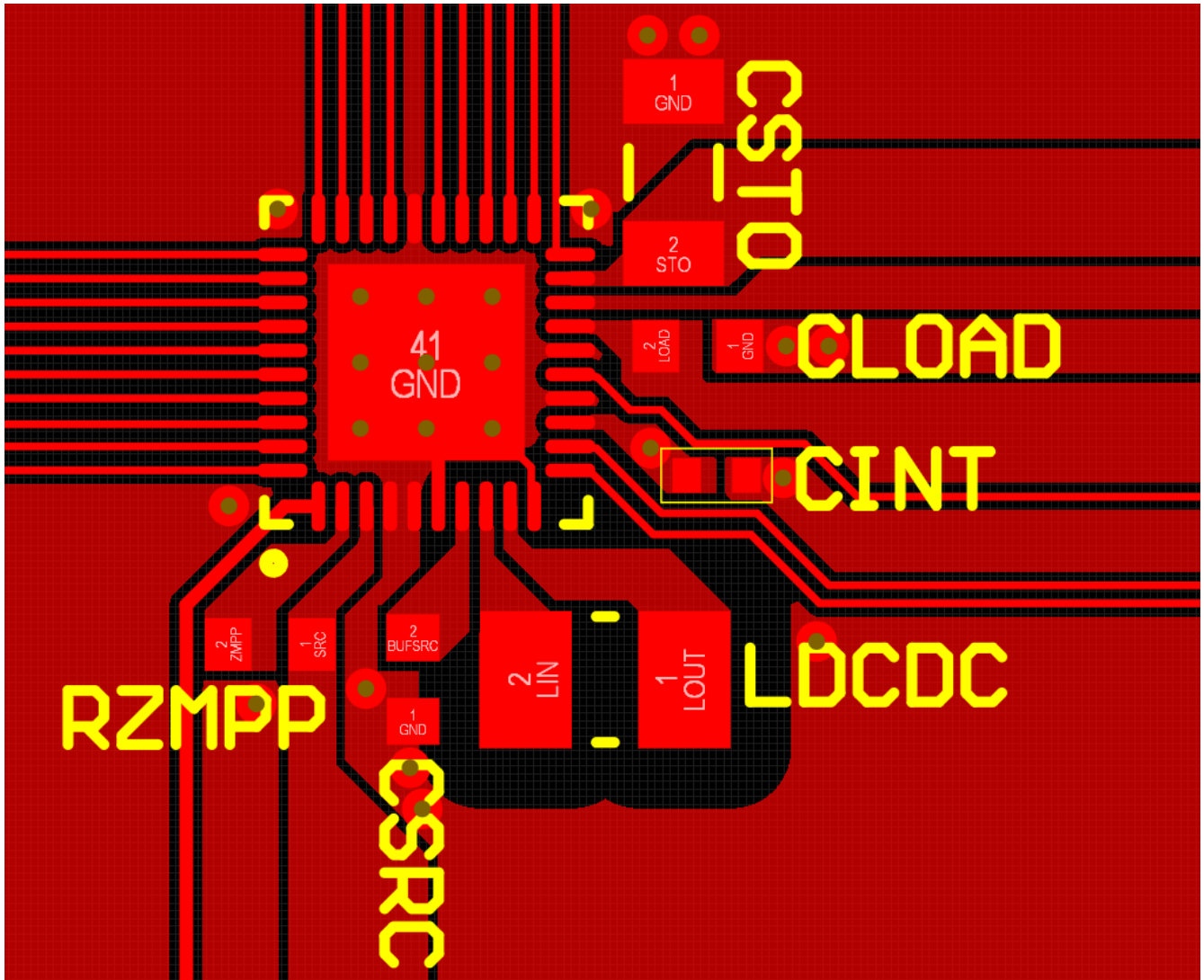


Figure 20: Layout Example for the AEM10330 and its Passive Components

NOTE: schematic, symbol and footprint for e-peas component can be ordered by contacting e-peas support team at support@e-peas.com

11. Package Information

11.1. Moisture Sensitivity Level

Parameter	Moisture sensitivity level (MSL)
QFN 40-pin - 5x5mm	Level 1, according to JEDEC-A113 standard

Table 14: Moisture sensitivity level

11.2. RoHS Compliance

e-peas product complies with RoHS requirement.

e-peas defines “RoHS” to mean that semiconductor end-products are compliant with RoHS regulation for all 10 RoHS substances.

This applies to silicon, die attached adhesive, gold wire bonding, lead frames, mold compound, and lead finish (pure tin).

11.3. REACH Compliance

The component and elements used by e-peas subcontractors to manufacture e-peas PMICs and devices are REACH compliant. For more detailed information, please contact e-peas sales team.

11.4. Tape and Reel Dimensions

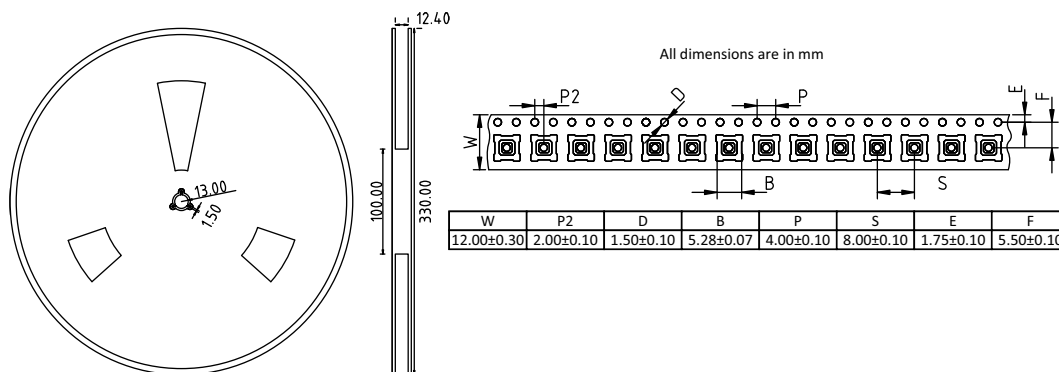


Figure 21: Tape and reel dimensions

11.5. Package Dimensions

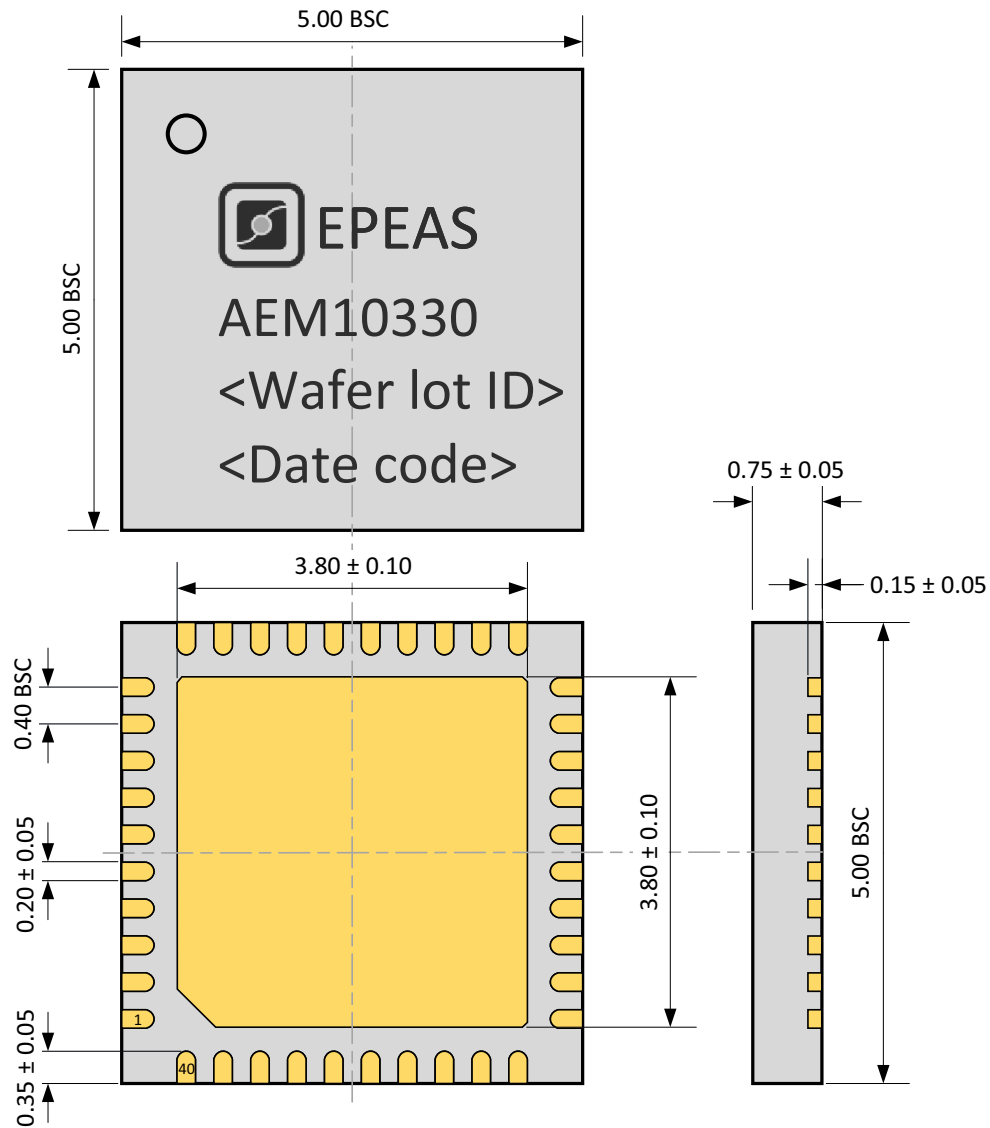


Figure 22: QFN 40-pin 5x5mm Drawing (All Dimensions in mm)

11.6. Board Layout

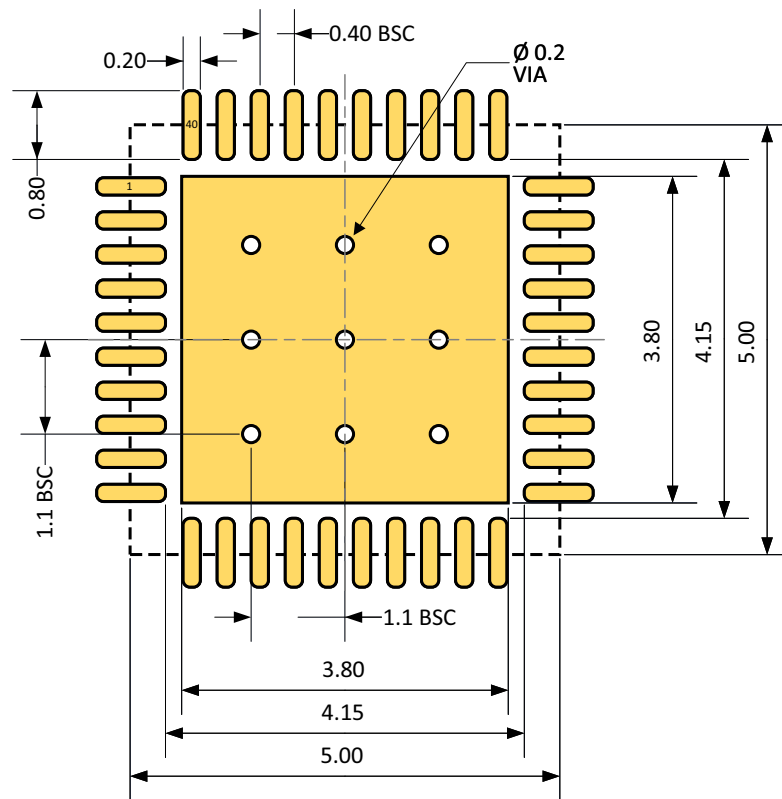


Figure 23: Recommended Board Layout for QFN40 package (All Dimensions in mm)

12. Glossary

C_{INT}

Decoupling capacitor on **VINT** pin.

C_{LOAD}

Decoupling capacitor on **LOAD** pin.

C_{SRC}

Decoupling capacitor on **BUFSRC** pin.

C_{STO}

Decoupling capacitor on **STO** pin.

L_{DCDC}

DCDC converter inductor.

$I_{LOAD,MAX}$

Maximum current available on **LOAD** pin.

I_Q

Quiescent current pulled from **STO** pin.

$P_{SRC,CS}$

Minimum source power required to coldstart the AEM10330.

MPP

Maximum Power Point.

MPPT

Maximum Power Point Tracking.

R_{MPPT}

For the DCDC converter, ratio between the open circuit voltage V_{OC} and the voltage regulation V_{MPP} determined by the MPPT (see Section 5.4 and Section 6.6).

R_{ZMPP}

Resistor used for ZMPP module, connected between **SRC** and **ZMPP** (see Section 6.7).

R_T

Sum of resistors to set the battery protection threshold voltages in custom mode. $R_T = R_1 + R_2 + R_3 + R_4$.

$T_{GPIO,MON}$

GPIO monitoring rate (except **EN_SLEEP** and **EN_STO_CH**).

$T_{LOAD,MON}$

LOAD monitoring rate.

$T_{MPPT,WAIT}$

Time during which the AEM10330 stops pulling current from **SRC** to let the source voltage rise to V_{OC} .

$T_{MPPT,MEASURE}$

Duration of the harvester open-circuit voltage (V_{OC}) measure.

$T_{MPPT,PERIOD}$

Period of the MPPT module V_{OC} evaluations (see Section 5.4 and Section 6.6).

$T_{EN_SLEEP,MON}$

EN_SLEEP pin monitoring rate.

$T_{STO,MON}$

STO monitoring rate.

V_{BAL}

Voltage on **BAL** pin.

$V_{BAL,MIN}$

Minimum voltage on the **BAL** pin for the balancing circuit to operate.

V_{OVDIS}

Minimum voltage accepted on the storage element before entering into **SHUTDOWN STATE** and resetting **ST_STO**.

V_{CHRDY}

Minimum voltage required on the storage element before asserting **ST_STO**.

V_{OVCH}

Maximum voltage accepted on the storage element before disabling its charging.

V_{VINT}

Voltage on **VINT** pin.

V_{LOAD}

Voltage on **LOAD** pin.

$V_{LOAD,MIN} / V_{LOAD,MID} / V_{LOAD,TYP} / V_{LOAD,MAX}$

Threshold voltages involved in the **LOAD** output voltage regulation. See Section 6.5 for further details.

V_{SRC}

Voltage on **SRC** pin.

$V_{SRC,CS}$

Minimum source input voltage required to coldstart the AEM10330.

V_{STO}

Voltage on **STO** pin.



V_{MPP}

Target regulation voltage of the DCDC **SRC** input, as determined by the MPPT module.

V_{OC}

Open-circuit voltage of the harvester connected to **SRC**.

13. Revision History

Revision	Date	Description
1.0	June, 2021	First version of the document
1.1	August, 2021	Minor modifications
1.2	March, 2023	- Various aesthetic improvements. - Explanations about BAL circuit. - Section with precisions about the use of CS_IN. - Removed LOAD_CFG[2] and related configurations. - New "behavior" oscilloscope graphs with improved description. - Moved various states description sections as sub-sections of a global section. - Supply State description: explanation about SRC being set to high impedance when all nodes are fully charged. - Updated "Typical use" of storage element vs STO_CFG[3:0] configuration. - Replaced "asserted/de-asserted" by "HIGH/LOW". - Changed CSRC from 15μF/0402 to 22μF/0603. - Updated "Recommended Operation Conditions" with minimum capacitor values including derating and tolerances. - Removed wrong dimension from package dimensions figure.
1.3	November, 2023	- Updated efficiency graphs. - Created section for pinout. - Updated schematics with new symbol. - Digital levels High/Low: replaced 0/1 notation by L/H. - Fixed example circuits errors. - Fixed typos and aesthetic issues. - Added layout guidelines with clearer layout examples.
1.4	February, 2024	Fixed wrong HIGH level of ST_STO in "Power and Status Pins" table.
1.5	July, 2026	- Reworked the whole datasheet. - Added REACH and RoHS compliances. - Added GPIOs, EN_SLEEP, LOAD, and STO monitoring rates. - Renamed the MPPT timings to $T_{MPPT, WAIT}$, $T_{MPPT, MEASURE}$, and $T_{MPPT, PERIOD}$. - Fixed confusion between R_{ZMPP} and R_{MPPT} in "Typical Electrical Characteristics" table. - Added $V_{BAL, MIN}$ and improved balancing circuit explanations. - Maximum load current graph moved from "System Configuration" section into "Performance Data" section. - Corrected "Typical Application Circuits" figures to match the descriptions. - Added cautionary statement along with storage element threshold voltages. - Added tape and reel dimensions. - Updated package and recommended board layout figures. - Added glossary.

Table 15: Revision History